University of Pavia
Faculty of Engineering
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Master Degree in Automation Engineering

Innovative Therapies for Hemispheric Brain Impairment: The Role of the PAL Robot in Cognitive and Motor Rehabilitation

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Abstract

Hemispheric brain damage, which can happen after a stroke, a traumatic injury, or a neurodegenerative disorder, makes motor and cognitive functions unevenly harmed. The old ways of doing rehabilitation have some problems. They relies too much on therapists, don't modify their treatment to each person, and can't easily change with each person's progress. This thesis answers the question, **How can socially assistive robotics be used to provide adaptive and autonomous neurorehabilitation for hemispheric brain damage?**

To find out more about this, the PAL Robotics TIAGO robot used and the ROS2 middleware to make and use a modular rehabilitation framework. The system has speech recognition, the ability to track facial expressions and body posture, the ability to navigate on its own, and robotic arm exercises that are controlled by the trajectory. Therapy routines are like **recipes** in that they are made to fit each person's needs. They change in real time depending on how the patient respons them.

The experimental simulations in a living-lab setting at the University of Pavia is done to see how well people could complete tasks, how responsive the system was, and how engaged users were through interaction. The results show that the robot can do multi-modal rehabilitation routines on its own with consistent accuracy and responsiveness. Data gathered through voice and visual feedback show that the system can keep an eye on therapy and change it as needed.

This work contributes a validated, ROS2-based robotic framework for neurorehabilitation that can be scaled up and used in an interactive way to support traditional therapy. The results set the stage for future progress in robotic care that is smart and focused on the patient.

Acknowledgments

This research would not be possible to complete sucessfully without the support and contributions of numerous individuals and institutions, to whom I express my deepest gratitude.

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This research was significantly helped by the open-source technologies from the **ROS2 (Robot Operating System 2)** framework and **PAL Robotics**. Their technological contributions to their development communities are adequately appreciated.

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Acronyms

LED Light Emitting Diode

UNIPV University of Pavia

Robolab Robotics Laboratory

ROS Robot Operating System

ROS2 Robot Operating System Version 2

SLAM Simultaneous Localization and Mapping

HRI Human-Robot Interaction

RGB-D Red Green Blue + Depth

TBI Traumatic Brain Injury

SAR Socially Assistive Robotics

CIMT Constraint-Induced Movement Therapy

RTT Repetitive Task Training

VR Virtual Reality

DOF Degrees of Freedom

AI Artificial Intelligence

XAI Explainable Artificial Intelligence

ECG Electrocardiogram

SLR Single Lens Reflex

ODF Open Document Format

SSH Secure Shell

CPU Central Processing Unit

GPU Graphics Processing Unit

API Application Programming Interface

GUI Graphical User Interface

NLP Natural Language Processing

PTSD Post-Traumatic Stress Disorder

ADHD Attention Deficit Hyperactivity Disorder

ASD Autism Spectrum Disorder

SQL Structured Query Language

ODS OpenDocument Spreadsheet

LiDAR Light Detection and Ranging

FPS Frames Per Second

AI Artificial Intelligence

Chapter 1

Introduction

"A design of any kind shows its real value when
taking beyond its original limits."

Tom Jennings

This chapter gives the clinical and technical background for this thesis, which looks at how robots can be used to help people with hemisphere brain damage recover and it starts by talking about how brain damage cause like stroke, traumatic brain injury, and neurodegenerative diseases affects the brain and the body movements. There are some problems with regular rehabilitation methods, so the goal is to make treatment plans that are more effective, flexible, and based on scientifically supported plans that include robotics. This work suggests a framework for robotic intervention by using the TIAGO robot's arm and base motion, LiDAR and camera sensing, and ability to interact. The ROS2 and Pal Web Interface framework will be used to build the system architecture that will be used to make the technical advances and experimental methods that will be talked about in the third chapter.

1.1 Reasons and Background

The brain has two halves, the left and right. Each half controls a different set of cognitive, motor, and emotional processes that are related to each other. People often think of the right hemisphere control the spatial awareness, visual-motor integration, and emotional processing. On the other hand, the left hemisphere is mostly use to control the language, logical reasoning, and processing things in order. Damage to either hemisphere can cause big problems with how the brain works, making it hard for someone to live alone and enjoy life.

The brain doesn't grow in a linear way. Studies in developmental neuroscience show that by the time a child is three years old, about 80% of the growth in the right hemisphere is already done. This early development allows the right hemisphere to take charge of emotional control, nonverbal communication, and early sensory integration in babies. Because of this, people who hurt the right side of their brain when they were young, whether it was from a stroke that happened before birth, birth defects, or trauma that happened early on, often have developmental problems that last for a long time. These include issues with controlling attention globally, communicating socially, and understanding space. Most of the time, these problems start when a person is young and last for the rest of their life.

But this study is mostly about adults who had brain damage on one side of their brain later in life and didn't have any major neurological issues when they were kids. When this happens, the damage is most often linked to conditions that were acquired, such as stroke, traumatic brain injury (TBI), tumors, infections, or neurodegenerative disorders. Adults are more likely to hurt their left hemisphere because it controls higher-level thinking and movement. Aphasia, trouble moving on the right side, and a lower ability to think are all common side effects.

This group of people has a harder time getting better. Adults who have been hurt have to relearn things they already knew, which can be hard because they don't have a lot of time and their brains are unstable or getting worse. This is not the same as cases that are present at birth or start early on. In the clinic, traditional therapeutic interventions often work well, but they take a lot of resources, need constant expert supervision, and can't be customized for each person.

In these situations, robotic systems seem like they could be a good way to help with recovery. But it can be hard to use robots to talk to people who have early-onset or severe developmental delays, both morally and technically. Using robots this way is dangerous because developmental problems can happen at any time, and people may not get what the robots are trying to say or be overstimulated by their cues. So, this thesis is mostly about adults, where operator know a lot about how their brains and behavior work and can safely use structured robotic protocols and test them in a consistent way.

The purpose of this study is to bridge the gap between traditional rehabilitation methods and the need for solutions that can be changed, are always available, and can be used by more people. The goal of this project is to find new ways for adults with acquired hemispheric brain damage to interact with robots, especially the TIAGO robot, to help them get their motor and cognitive skills back. This sets the stage for making and testing screened intelligent robotic systems based on ROS2 that can offer targeted therapeutic interventions.

1.2 What are Causes of Hemispheric Brain Damage

Hemispheric brain injury can occur from several medical diseases and physical injuries that mostly impact one hemisphere of the brain. These causes can occur suddenly, as a result of trauma or stroke, or progress gradually due to existing neurological conditions. Knowing the many sources of hemispheric injury is crucial for identifying suitable rehabilitation treatments. The following are the primary groups of circumstances recognized to cause such damage.

Stroke

Stroke is one of the most common and dangerous ways for adults to hurt one side of their brain. When blood flow to the brain is stopped or blocked, it cuts off oxygen and nutrients to the area that is affected. This can damage nerve tissue permanently and make the brain work less well. There are two main types of strokes:

1. **Ischemic Stroke:** This happens when a blood vessel gets blocked, usually by a clot. This is the most common type, and it can hurt certain parts of the brain.

2. **Hemorrhagic Stroke:** A hemorrhagic stroke happens when a blood vessel breaks and blood leaks into or around the brain. This makes the pressure go up and hurts the brain tissue around it.

The functional effects of a stroke depend a lot on which hemisphere is affected as shown in Figure 1.1:

1. **Damage to the left hemisphere:** This can cause weakness or paralysis on the right side, trouble with speech and language (aphasia), and trouble with logical reasoning.

2. **Damage to the right hemisphere:** usually causes problems with movement on the left side, problems with spatial awareness, ignoring the left side, and trouble processing emotions.

Structured robotic help can make a big difference in the targeted and intensive rehabilitation that stroke-related brain injuries need.

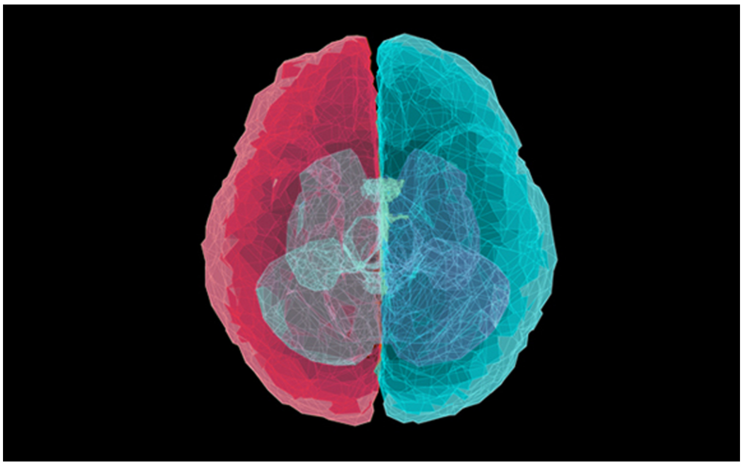


Figure 1.1: Illustration of Stroke-Induced Hemisphere Damage, adapted from [3]

Traumatic Brain Injury (TBI)

Traumatic Brain Injury (TBI) is damage to the brain's structure or function that is caused by outside mechanical forces. Motor vehicle accidents, falls, sports-related impacts, or physical attacks are common causes of these kinds of injuries. The severity and effects of TBI depend a lot on the type, strength, and location of the impact, as well as the person's age and overall health.

TBI can cause either focal or diffuse brain damage. Localized injuries happen at the point of impact or on the opposite side because of a "contrecoup" effect. Widespread injuries happen because of rapid acceleration and deceleration forces.

When TBI affects one side of the brain as shown in the Figure 1.2, the problems that come up are often similar to those that happen with other localized brain injuries:

1. **Damage to the left hemisphere:** Damage to the left hemisphere can make it hard to speak and understand language (like aphasia), read and write, and think logically and analytically.

2. **Damage to the right hemisphere:** Damage to the right hemisphere usually leads to problems with visuospatial processing, attention, recognizing and controlling emotions, and ignoring the left side of the body or space.

Because TBI is so complicated and hard to predict, rehabilitation usually needs a team of experts from different fields. Robotic systems look like they could be useful for structured motor retraining, objective assessment, and cognitive engagement in recovery after a TBI.

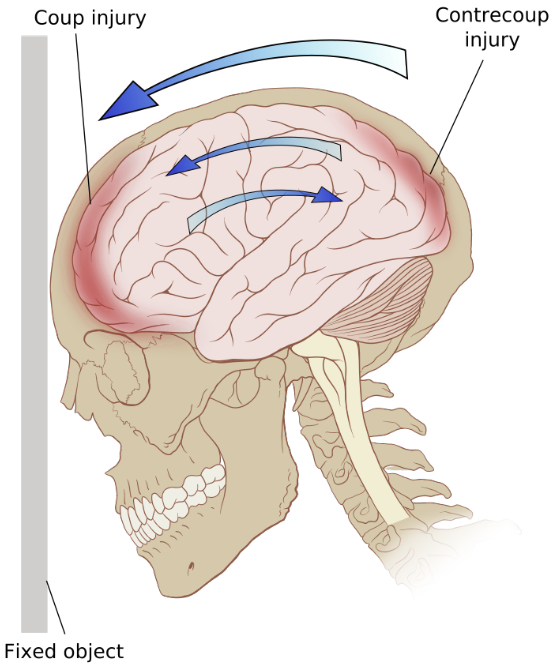


Figure 1.2: Schematic representation of contrecoup injury in Traumatic Brain Injury (TBI) [35]

Brain Disorders That Are Present at Birth

Congenital brain disorders are neurological problems that happen before or during birth. Birth asphyxia, infections in the womb, exposure to toxins before birth, or strokes during or after birth are some of the things that can cause these conditions. The brain damage can stop normal neural development, which often leads to long-term neurological problems like cerebral palsy, intellectual disabilities, or developmental delays. These problems usually show up early in life and last for the rest of a person's life. This makes rehabilitation very difficult, especially because the brain is changing in ways that make it more adaptable during early development.

Tumors in the Brain

Brain tumors, which can be either malignant or benign, can cause neurological problems by compressing or invading nearby brain tissue and interfering with its normal functions. The way symptoms show up in a patient depends a lot on where the tumor is, how big it is, and how fast it grows. The location in the hemisphere is very important:

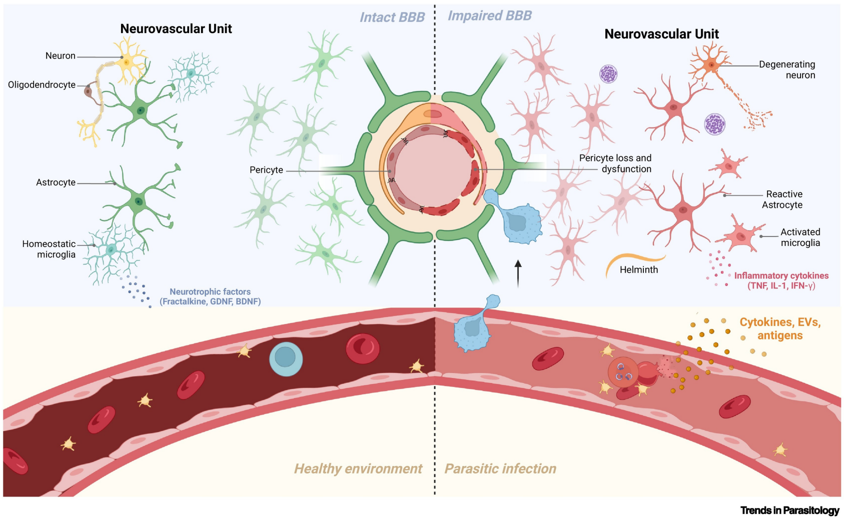


Figure 1.3: Inflammatory and Infectious Processes Affecting Brain Tissue [34]

- **Left Hemisphere Involvement:** Patients often have trouble with memory, aphasia, or other language-related issues, as well as problems with logical thinking.
- **Right Hemisphere Involvement:** Common signs include confusion, bad decisions, trouble with visuospatial processing, and problems with creative and emotional functions.

Because brain tumors grow over time, they need to be diagnosed and treated quickly, with rehabilitation plans that are specific to the cognitive and motor problems caused by the tumor and its treatment.

Infections and Swelling

Infections and inflammatory(swelling) shown in Figure 1.3, diseases of the central nervous system are major causes of brain damage in one hemisphere. Viral and bacterial encephalitis, meningitis, and autoimmune disorders like multiple sclerosis cause inflammation that can damage neural tissue in either hemisphere, depending on where and how bad the disease is. These inflammatory processes often cause neuronal loss, demyelination, and synaptic connectivity problems, which show up as cognitive deficits, motor problems, seizures, or changes in behavior.

The fact that these conditions are so different makes it hard to diagnose and treat them because the severity and location of the inflammation have a big effect on the symptoms and the success of rehabilitation. Early detection and targeted inter-

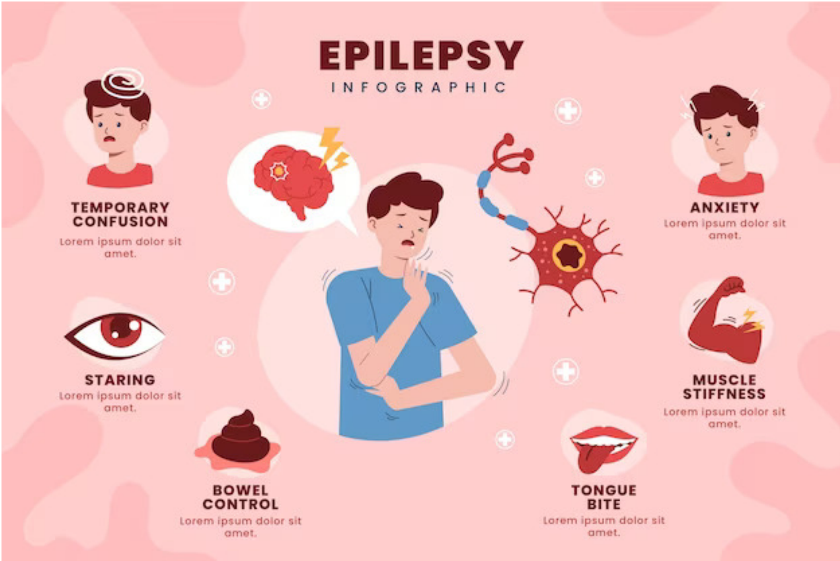


Figure 1.4: Illustration of Epileptic Seizure Activity and Brain Regions Affected [27]

vention are very important for reducing long-term neurological effects and speeding up functional recovery.

Epilepsy and Seizure Disorders

Epilepsy is a neurological disorder that causes seizures that happen over and over again without any known cause. These seizures are caused by abnormal electrical activity in the brain. Some types of epilepsy only affect one hemisphere of the brain, which can cause focal neurological deficits depending on where the problem is. Seizures that happen over and over again can damage the structure and function of the hemisphere that is affected as shown some symptoms in Figure 1.4.

If drugs don't work to stop seizures (medically refractory epilepsy), surgery like lobectomy or hemispherectomy may be necessary. These procedures do lower the number of seizures, but they also change how the affected hemisphere works, which makes rehabilitation harder, especially when it comes to keeping or making up for lost motor, cognitive, and sensory functions.

Managing and rehabilitating people with epilepsy requires a multidisciplinary approach, and advanced robotic systems may be useful for personalized therapy and functional assessments as shown some symptoms in Figure 1.5.

Figure 1.5: Seizure Manifestations and Hemispheric Implications [13]

Neurodegenerative Diseases.

Neurodegenerative diseases are a large group of disorders that cause neurons to lose their structure or function over time as shown the growth in Figure 1.6. This often leads to permanent problems with thinking, moving, and behaving. Alzheimer's disease and frontotemporal dementia are two examples of conditions that often start out asymmetrically, with neurodegeneration affecting one hemisphere of the brain more than the other at first.

This lateralized degeneration causes hemisphere-specific problems. For example, involvement of the dominant hemisphere can cause aphasia and language problems, while deterioration of the non-dominant hemisphere can make it harder to control emotions and see things in three dimensions. As the disease gets worse, it affects both sides of the brain, which makes the global decline in neurological function worse.

To make targeted rehabilitation plans, it's important to know how progression patterns differ between hemispheres. Robotic-assisted interventions could help reduce functional losses by making it easier to do repetitive, task-specific training that targets the deficits in the affected hemisphere. This is possible because they use sensor-driven feedback and adaptive control.

1.3 Problems with Diagnosis and Rehabilitation

Conventional rehabilitation methods for hemispheric brain damage have a lot of problems that make them less effective and harder to get to around the world.

First, there is a lot of variation in the quality and availability of healthcare.

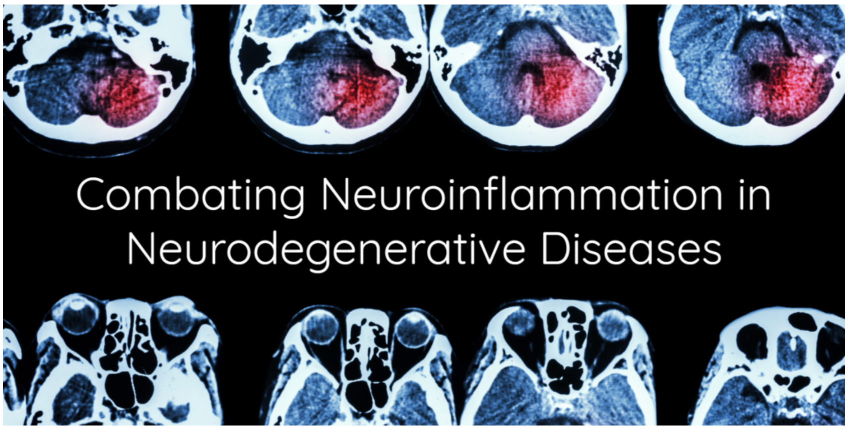


Figure 1.6: Illustration of Neurodegenerative Disease Progression and Hemispheric Impact [30]

For example, rural and underserved areas often don't have access to specialized rehabilitation services or trained staff. This difference makes it harder to get help and makes things worse.

Second, keeping patients motivated over the long and difficult course of therapy is hard because emotional and cognitive problems can make them angry, sad, and less interested. Traditional treatments often don't offer the interesting, tailored approaches that are needed to keep people on track.

Third, the different ways that brain injuries happen and the fact that each person's brain is different means that recovery can happen in very different ways. This means that treatment plans need to be flexible and adaptable, which is hard for traditional static protocols to do.

Rehabilitation is very demanding on therapists, which can lead to fatigue, burnout, and less availability, especially in health systems that are in high demand. Also, most of the assessment tools available now only use qualitative or semi-quantitative measures. They don't have the ability to do continuous, objective, and multimodal evaluations, which are very important for making small changes to therapy.

Another big problem is that therapy sessions need to be able to change in real time to account for changes in a patient's fatigue, attention, and cognitive load. Therapy may not work as well if it doesn't respond in real time. Also, for rehabilitation to work, medical and therapeutic teams need to work together, which is often hard because of communication problems.

Economic and policy limitations make rehabilitation shorter and less intense, especially in places with few resources. Lastly, patients and doctors have very dif-

Figure 1.7: Robotic platforms enhancing motor and cognitive rehabilitation through interactive and adaptive therapies. [33]

ferent levels of technological literacy and acceptance, which makes it hard to use advanced robotic systems.

Robotic platforms can provide innovative, scalable, and personalized rehabilitation solutions that can help with these complex problems. These solutions can be achieved through intelligent, adaptive, and culturally aware interactions.

1.4 Robotic Intervention for Hemispheric Damage

Recent progress in robotics opens up new possibilities for overcoming many of the problems that come with traditional neurorehabilitation. Robotic systems can be easily added to both diagnostic and therapeutic frameworks, as indicated in Figure 1.7 making treatment plans more accurate, repeatable, and tailored to each patient. Some important uses are:

1. **Motor Rehabilitation:** Robotic exoskeletons and other assistive devices help people with strokes or traumatic brain injuries make guided, repetitive limb movements that are important for neuroplastic recovery. These devices can accurately control the paths of movement and resistance, which helps people relearn how to move with a lot of consistency.

2. **Cognitive Assessment:** Socially assistive robots with multimodal sensors (like speech recognition and cameras) can dynamically test cognitive functions like memory, attention, emotional responses, and verbal fluency. These robots are objective and can be used to test a wide range of people.

3. **Speech Therapy:** Robotic platforms with Natural Language Processing (NLP) can do personalized speech and language exercises for people with aphasia and other communication problems. They can change in real time based on how the patient responds and how well they are doing.

4. **Personalized Treatment:** Robotic systems use data from sensors to keep an eye on how well a patient is doing and how tired they are at all times. They then change the intensity and complexity of the therapy to keep the patient interested and get the best results.

5. **Remote Monitoring and Tele-rehabilitation:** Wireless technology lets clinicians continuously collect and send patient data, making it possible to supervise patients from afar, keep track of their progress, and make timely changes to their treatment plans. This means that rehabilitation can happen outside of clinical settings as well.

Some well-known examples are robotic platforms like MIT Manus, which is widely used for upper limb motor rehabilitation, and humanoid robots like NAO and Pepper, which are used as interactive agents in cognitive and emotional therapies. These technologies show how robotics, neuroscience, and clinical rehabilitation can come together to create new ways to care for patients that are tailored to their needs.

1.5 Purpose of the Thesis and Why?

Hemispheric brain damage is a major clinical problem around the world because it happens so often and causes a wide range of complex and often debilitating problems. Traditional rehabilitation methods often have trouble providing therapy that is tailored to each patient's unique recovery path and is consistent, personalized, and adaptable. These problems can lead to longer recovery times, different results, and a lot of stress on healthcare resources.

This thesis is based on the need to focuse and create robotic interventions that combine motor, cognitive, and speech rehabilitation in a way that works for people with hemispheric brain damage. The main goal is to use the PAL Robotics TIAGO platform as a testbed to design, build, and test an intelligent robotic system that can interact with people in a way that adapts to their needs and deliver personalized therapy.

This work includes creating personalized motion sequences (recipes) that are in line with therapeutic goals, combining vision and speech processing modules for real-time patient monitoring, and using adaptive algorithms to change the intensity of rehabilitation based on how the patient responds. The system will be test to see if it is effective at getting patients involved, helping them recover, and giving useful clinical data for making changes to their therapy. This research aims to add scalable, culturally aware, and user-centered solutions that improve neurorehabilitation outcomes and make it easier for more people to get high-quality care by connecting robotics, neuroscience, and clinical rehabilitation.

1.6 Research Implementation and Practical Setup

The TIAGO robot, which was made by PAL Robotics, is used as the experimental platform for this study. TIAGO is a humanoid robot that helps people with social tasks and can do complicated tasks on its own. TIAGO is currently at the UniPv Living Lab, where it is the main tool for creating, carrying out, and testing rehabilitation scenarios that are specifically designed for patients with hemispheric brain damage.

The system uses TIAGO's advanced hardware shown in Figure 1.8, such as a 6-degree-of-freedom robotic arm that can control motors very precisely, an RGB-D camera for rich visual sensing, and a 2D LIDAR sensor for finding its way around. There are also built-in modules for autonomous navigation and face detection that help the robot interact with patients in changing environments.

The Robot Operating System 2 (ROS2) framework is used to build the software architecture. Within this framework, custom control nodes, called "recipes," are programmed to carry out certain movement sequences. These sequences are divided into several levels, each of which is meant to gradually test and improve patients' motor skills and cognitive responses. This structured approach makes it easier to engage with patients in a consistent way and lets therapy change as the patient gets better.

The TIAGO system also includes multimodal sensory feedback by constantly watching how patients react through vision-based analysis of their facial expressions, gaze direction, and body posture. This information is used in real time to check how engaged the patient is and how well they move. There is also a speech detection module built in so that patients can talk to the robot or give it commands, which makes the interaction between humans and robots more natural.

Figure 1.8: TIAGO Robot from PAL Robotics deployed for neurorehabilitation research. [19]

This platform shows that socially assistive robots can be used not only as therapeutic tools but also as active participants in patient assessment and personalized rehabilitation. It does this by combining precise robotic actuation, advanced perception capabilities, and adaptive control. This integration makes it possible to create more effective, scalable, and data-driven neurorehabilitation protocols that are specific to the needs of patients with hemispheric brain damage.

TIAGO moves the patient's arms and body in a set way during each rehabilitation session while also keeping an eye on how the patient is responding with its built-in RGB-D vision system. This vision subsystem looks at facial expressions, gaze direction, and body posture in real time to see how engaged and responsive the patient is. An integrated speech recognition module also makes it possible for patients to talk to each other and give commands that are understood by clinicians or the robotic system.

TIAGO is an active part of the rehabilitation process because it combines motor control, vision, and speech inputs. It helps with therapeutic activities and collects clinically relevant data at the same time. The successful use of this robotic setup proves that humanoid robots can be used not only as therapy assistants but also as ways to keep an eye on patients all the time and adapt their rehabilitation to their needs using the component and Functionality offer by TIAGO Table 1.1.

Table 1.1: TIAGO Robot Configuration for Neurorehabilitation

Component	**Functionality in Rehabilitation**
6-Axis Robotic Arm	Executes guided motor tasks; facilitates upper limb coordination training and evaluation
RGB-D Camera	Captures patient facial expressions, gaze, and posture to assess engagement and motor responses
2D LIDAR Sensor	Enables autonomous navigation to predetermined test locations within the environment
Face Detection Module	Detects and tracks patient attention to support adaptive interaction during tasks
ROS2 Control Nodes	Custom-developed software modules managing movement recipes and system behavior
Speech Detection Module	Provides voice-based interaction capabilities for patient and clinician communication
Navigation Stack	Supports autonomous movement of the robot within the experimental setting
Level-Based Movement Recipes	Hierarchically structured motion sequences to progressively challenge and evaluate patients
Data Collection Pipeline	Aggregates and logs visual and speech data for offline analysis and therapy adaptation

1.7 Contribution of the Thesis

This thesis makes important progress in the development and use of the PAL TIAGO robot as a semi-autonomous platform for neurorehabilitation. The main contributions are as follows:

1. **Integrated ROS2-Based Architecture:** A new, modular software framework that brings together multi-modal human-robot interaction, autonomous navigation, arm trajectory control, face recognition, and session data management into a smooth, real-time system that can be used in clinical settings.

2. **Robust Environment Mapping and Navigation:** Using LiDAR data and SLAM, along with user-defined navigation waypoints, to create a semi-manual, flexible 2D mapping and localization system that lets robots move safely and accurately in rehabilitation spaces.

3. **Offline Voice Command Interface:** Implemented an offline speech recognition pipeline using the open-source Vosk toolkit. This lets patients and therapists give commands in natural language without needing the internet, making it easier to use in clinical settings.

4. **Personalized User Interaction via Face Recognition:** OpenCV is used to integrate real-time face detection and recognition with a local identity database. This lets the robot customize therapy sessions and keep track of how engaged the patient is.

5. **Predefined Therapeutic Arm Movements:** Use ROS2 action interfaces to design and carry out 16 therapeutic arm trajectories. This makes it possible to create custom therapy routines and repeatable motion sequences that are important for consistent neurorehabilitation exercises.

6. **Full Data Logging and Visualization:** Setting up a structured data pipeline using SQLite and ODS formats to keep track of session metrics, performance data, and user interactions, along with visualization tools to help with clinical evaluation and decision-making.

7. **Safety and Fault Management:** To make sure that patients are safe and the system is strong while it is running on its own, multiple layers of safety mechanisms are put in place, such as emergency stop commands, fault recovery, and watchdog timers.

8. **System Validation and Performance Analysis:** A thorough empirical evaluation of each system component using both simulation and controlled real-world tests done on the author (AK). The full integrated model hasn't been used in clinical settings yet, but these section-by-section tests show that it has high trajectory accuracy, strong and responsive offline voice command recognition, and reliable emergency stop functionality. This sets a strong foundation for future full-scale use.

Together, these contributions improve robotic neurorehabilitation by making a clinically useful, flexible, and safe robotic assistant that can improve therapy delivery and patient engagement.

1.8 Outline of the Thesis

There are six chapters in this thesis, each one focusing on a different part of the research and helping to reach the main goal of creating an adaptive robotic system for neurorehabilitation with the TIAGO robot. The outline below gives a short summary of what each chapter is about and what it is meant to do.

Chapter 1: Introduction gives the clinical background and reasons for doing the study. It talks about what causes and what happens when one hemisphere of the brain is damaged, the problems with traditional rehabilitation methods, and how robotic systems are becoming more important in therapy. It also talks about the research's goals and scope, as well as the environment in which it will be carried out and the system's goals.

Chapter 2: Literature Review looks at research that is relevant to neurorehabilitation, socially assistive robotics, SLAM and navigation, human-robot interaction, computer vision for recognizing posture and emotion, and controlling robotic arms. This chapter talks about the problems with current solutions and the research gap that this thesis wants to fill with a ROS2-based integrated system.

Chapter 3: Tools and Frameworks talks about the hardware and software that were used in this study. It goes into detail about how to set up the TIAGO robot's connectivity, how to use the web programming interface, and the parts that make the robot able to navigate, process sensor data, and interact with people.

Chapter 4: Implementation talks about the modular software architecture that was made for the rehabilitation framework. It includes starting up the system, the speech and web interface, face recognition, navigation, robotic arm control, therapy recipe logic, safety features, and data management. Shows about how each module is designed and built in relation to its role in the rehabilitation workflow.

Chapter 5: Results shows how the developed system was tested and how well it worked. The chapter talks about the results of specific tasks, like how well the person could navigate, how their arms moved, and how well they could do motor tasks based on a recipe. The framework's reliability and effectiveness in a lab setting are also shown through comparative analysis and error evaluation.

Chapter 6: Conclusions brings together the research results and talks about how the system has helped the field of intelligent, adaptive robotic rehabilitation move forward. The chapter also talks about the chapter's limitations and suggests ways to move forward, such as using AI to personalize things, affective computing, cloud integration, and clinical deployment strategies.

Chapter 2

Literature Review

"The wireless telegraph is not difficult to understand..."

Albert Einstein

This chapter brings together basic and cutting-edge research that is shaping the crossroads of robotics, artificial intelligence, and neurorehabilitation. It starts by looking at the neurological causes of hemispheric injury, focusing on the many cognitive and motor problems that can happen with unilateral brain injuries and the problems with traditional rehabilitation methods. The conversation then moves on to new therapies that use technology to help people. It focuses on how Socially Assistive Robotics (SAR) fills in the gaps in individualized, adaptive care by using multimodal feedback and affective computing.

Key technical pillars, like SLAM-based navigation, computer vision for analyzing patient interaction, and ROS2-driven robotic control, are carefully examined. This is done with help from research on 6-DOF arm manipulation and smooth motion. The chapter ends by listing unmet needs, such as the need for integrated motor-cognitive frameworks and dynamic patient engagement measures. It says that a ROS2-based SAR system that uses real-time adaptability and tasks that are given to each hemisphere will greatly improve the results of rehabilitation.

2.1 The Neurological Basis of Hemispheric Damage

Cerebral lateralization, also known as hemispheric specialization, is a well-known pattern in the human brain. The left hemisphere is mostly in charge of language processing, analytical reasoning, math skills, and fine motor skills. The right hemisphere, on the other hand, is linked to visuospatial awareness, recognizing emotions, processing information as a whole, and controlling attention [9, 29]. This difference is very important in figuring out what kind of problems and how bad they are after a one-sided brain injury, especially a stroke.

Clinical research shows that damage to one hemisphere has a big effect on how the brain works. For example, lesions in the right hemisphere are strongly linked to spatial neglect and problems with attention. A meta-analysis says that about 73% of people who survive a stroke in the right hemisphere have unilateral spatial neglect, which makes it very hard for them to see or pay attention to things on the other side of space [2]. This condition makes it harder to do everyday things and makes rehabilitation less effective.

Another common result of hemispheric damage is motor problems, which can affect both gross and fine motor control. Research shows that almost half of stroke patients still have major motor problems six months after the event, even though they have received standard treatments [11]. These problems often include hemiparesis, apraxia, and loss of coordination, which make it much harder to live independently and enjoy life.

Also, there is evidence that traditional neurorehabilitation methods may not work well in the long term for people with chronic neurological deficits. After a few months, many patients stop getting better, and it becomes hard to keep making progress without adaptive or intensive treatment [6]. This shows a major flaw in the way therapy is done now.

Because of these problems, there is a growing need for new rehabilitation methods that are tailored to each patient. Robotic systems, especially those that combine cognitive feedback and adaptive help, are a promising direction. These kinds of systems can be set up to send stimuli to specific hemispheres, change based on how well each patient is doing in real time, and encourage neuroplasticity through repetitive, task-specific training. This personalized, data-driven approach is very important for helping people with hemispheric brain damage get better and stay independent in the long term.

2.2 Traditional vs. Technology-Assisted Rehabilitation

Physical therapy, occupational therapy, and speech-language therapy are all traditional ways to help people recover from brain injuries. They have been the mainstay of post-stroke and neurological recovery for a long time [8]. These traditional methods have been shown to work in the clinic to improve motor and cognitive functions, but they are often limited by a number of inherent problems. First, therapy sessions are hard to get to and not very intense because there aren't enough therapists or resources available, especially in places that don't have a lot of them [16]. Second, subjective clinical scales and observational assessments are often used to measure outcomes, which can make it hard to fairly judge a patient's progress [10]. Third, it's hard to get the high repetition and task-specific intensity needed to drive neuroplasticity and functional gains in traditional frameworks, which leads to less than ideal long-term recovery outcomes [18].

On the other hand, technology-assisted rehabilitation platforms, such as robotic-assisted devices, virtual reality (VR), and computer-based cognitive training, have many benefits that help get around these problems. These systems make it possible to consistently deliver high-intensity, repetitive therapy while also giving objective, measurable metrics for tracking progress [12]. The differences in clinical outcomes from different types of rehabilitation. It shows that technology-assisted interventions lead to better motor and cognitive gains than manual therapy.

Even though these systems have some good points, most of the technology-based rehabilitation systems that are already out there tend to separate motor and cognitive domains instead of combining them into a single therapeutic framework. This limitation makes it harder to fully treat complex neurological impairments, which often show up as both motor and cognitive problems at the same time [37]. As a result, more and more researchers in the field of rehabilitation agree that future therapeutic platforms should focus on multimodal integration, which means combining motor practice with cognitive engagement to get the most out of neuroplastic adaptation and functional recovery [26].

To make these kinds of integrated systems, need to improve sensor technologies, real-time data processing, and adaptive algorithms that can tailor therapy to each patient's needs. These new ideas will help fill in the gaps in how well and easily rehabilitation works, which will make life better for people with hemispheric brain damage and other neurological disorders.

2.3 Socially Assistive Robotics (SAR) in Therapy

Socially Assistive Robotics (SAR) is a field that combines robotics, human-computer interaction, affective computing, and rehabilitation science to make smart systems that can help people heal through social interaction instead of just physical manipulation [5]. These robots can be friends, coaches, or mediators who motivate and help patients with their rehabilitation tasks while also reading their emotions in real time. The basic idea is that emotional engagement, when brought about through social interaction, can improve user compliance, motivation, and, in the end, therapeutic outcomes.

SAR platforms have a lot of important functional features for neurorehabilitation. First, a lot of systems have real-time task adaptation algorithms that change the difficulty and order of tasks based on how well the patient is doing, making sure that the therapy is tailored to each person [22]. Second, using multiple types of biofeedback, such as sight, sound, and touch, has been shown to keep users interested and help them learn sensorimotor skills (Kim, 2023). Third, advanced sensing technologies like facial expression recognition, eye gaze tracking, and motion capture make it possible to keep an eye on a patient's engagement, emotional state, and physical behavior all the time [25].

Affective computing, personalization, and adaptive feedback are the three main parts of modern SAR frameworks. shows how these parts fit together. Our system builds on these ideas by adding hemisphere-specific task modules that are designed to help people with cognitive-motor deficits that are common after unilateral brain injuries. Also, adding physiological engagement metrics like heart rate variability or pupil dilation gives an objective way to measure how responsive a user is, which makes the rehabilitation experience more adaptable and complete.

SAR is a powerful way to help people with neurological problems in the future, especially when it is designed to meet the specific needs of people with different types of neurological problems.

2.4 Path Planning and Navigation in Therapeutic Settings

Mobile assistive robots that work in clinical and therapeutic settings must be able to navigate accurately and on their own. Hospitals, rehab centers, and patients' homes are always changing and full of people, so they need strong perception, localization, and trajectory planning systems to keep everyone safe, adaptable, and getting the care they need.

2D LiDAR-Based SLAM: Using 2D LiDAR sensors for Simultaneous Localization and Mapping (SLAM) is one of the most advanced ways to navigate indoors. GMapping and Cartographer are two algorithms that use probabilistic methods to build occupancy grid maps step by step while also estimating the robot's pose [4].

These two-dimensional maps work very well in small indoor spaces and are the basis for accurate navigation and location. Because they don't need a lot of processing power, they are perfect for use in medical robots that need to respond quickly.

RGB-D Sensor Fusion for Better SLAM: To get around the problems with 2D images, many modern therapeutic robots use RGB-D sensors that record both color and depth data. Depth-enhanced SLAM frameworks, like RTAB-Map, go beyond traditional mapping by making 3D or semantically rich 2.5D maps that help with obstacle avoidance and spatial reasoning in complicated, unstructured places [7]. Combining LiDAR and RGB-D makes object recognition and mapping more reliable, especially in places with different lighting and obstructions, like rehabilitation gyms and home therapy spaces.

Path planning and autonomous navigation: After a reliable map is made, path planning is very important for robots that move on their own. The Navigation Stack (Nav2) in ROS2 uses global planning algorithms like A* and Dijkstra, along with dynamic local planners and costmaps that show current sensor input to avoid people and obstacles in real time [14]. This ability to react in real time is very important in therapeutic settings, where patients may move around in ways that are hard to predict or use assistive devices. Nav2 also has dynamic obstacle tracking, multi-layered costmaps (for things like inflation and social distancing), and recovery behaviors to make sure that service is never interrupted.

TIAGO and other socially assistive robots can use SLAM and navigation technologies to move around on their own between patient rooms, therapy stations, or shared clinic hallways. This mobility not only helps with logistical tasks like delivering materials or helping therapists, but it also allows for interactive, task-oriented therapy sessions where the robot moves around based on patient cues or specific rehabilitation tasks. By adding SLAM and real-time planning to its autonomy stack, the robot becomes a therapeutic agent that is aware of its surroundings and can easily fit into changing clinical workflows.

2.5 Computer vision for finding faces and analyzing posture

RGB-D cameras and other computer vision technologies have become very important for making robots able to interact with people naturally and quickly in therapeutic settings. Combining visual and depth data helps us understand how people feel and how their bodies work in more than one way. This lets us create personalized engagement strategies that are very important in neurorehabilitation.

Face detection is a basic part of this field that lets the robot find and follow the user's face so that they can interact with each other. OpenCV and DLib are two well-known computer vision libraries that modern systems use. They use machine learning and feature extraction methods to find facial landmarks in real time. These facial features, like eyebrows, eyes, nose, and mouth contours, make it possible to recognize emotions in real time by looking at how facial muscles move in a geometric way [36]. It is important to be able to read a patient's feelings in order to change the tone, speed, and difficulty of therapeutic tasks based on how open the user is to them.

In addition to detecting emotions, estimating gaze and tracking head pose are two other important ways to measure how engaged and focused a patient is. The robot can tell if a patient is focused on the task or not by looking at the direction of their eye movement and head position. This lets the system change its prompts or stimuli as needed. This not only makes the interaction more consistent, but it also lets the robot and patient get to know each other better on a social level.

Depth-assisted posture and skeletal tracking are especially useful for physical rehabilitation. Shotton et al. proposed Kinect-based body tracking systems for gaming platforms that have been changed for use in clinics to keep an eye on joint trajectories and limb alignment in real time [28]. These frameworks let therapists and robots check to see if the prescribed movements are being done correctly, find any deviations or compensatory patterns, and give feedback right away. This kind of monitoring is very important for helping people learn how to move and keeping them from getting hurt during rehabilitation sessions where they are not being watched.

Combining facial affect recognition with posture analysis gives a complete picture of how a patient behaves, which lets therapy systems change the content and intensity of their sessions in real time. These computer vision technologies not only help with precise rehabilitation, but they also boost user motivation and engagement, which are both important for long-term recovery in neurological patients.

2.6 Voice Recognition for People and Robots Interaction

Voice recognition technology is a key part of natural and intuitive human-robot interaction, especially in therapeutic settings where ease of use and accessibility are very important. Recent improvements in offline speech recognition frameworks like Vosk and DeepSpeech make it possible to process spoken commands locally, which protects patient privacy and reliability without needing to be connected to the cloud [32]. These systems turn natural language input into separate commands that work perfectly with robot control architectures, which are usually built using ROS2 nodes and topics. This integration lets robots understand spoken commands and turn them into actions or speech that are appropriate for the situation.

Advanced human-robot interaction frameworks use contextual information to change the robot's verbal responses on the fly, going beyond just basic command execution. This includes changing the tone, giving positive feedback, or giving corrective advice based on a real-time assessment of how well the patient is doing their tasks and how they are feeling [31]. Robots can keep users interested and motivated while reducing cognitive overload or frustration by changing the way they interact with them in this way. In rehabilitation settings, context-aware dialogue systems are especially useful because they can help therapists by keeping patients focused and following therapy protocols.

In general, combining offline voice recognition with smart dialog management makes assistive robotic systems easier to use. This not only lets patients communicate naturally without the need for complicated input devices, but it also takes some of the work off of healthcare providers, allowing for more personalized and scalable therapy delivery.

2.7 6-Axis Robotic Arm Control

The University of Pavia imparted knowledge beyond the use of robotic arms; it also instructed me in ensuring their safe interaction with individuals. Envision a 6-DOF robotic arm maneuvering around a patient during therapy, adapting its movements in real time. The efficacy of real-time torque and proximity detection is remarkable. The strength and sensitivity of a robot must be balanced in rehabilitation.

One of the most interesting things, have learned was how to make a robotic arm move smoothly using something called joint-space trajectory blending. In simple terms, it means making the robot move from one point to another without sudden or jerky motions. This isn't just about looking smooth-it's also about safety and comfort. A sudden movement could scare someone or even cause harm. In class and lab sessions, worked on creating these smooth motions so the robot could interact with people in a calm and natural way.

What really made a difference was learning to use ROS2. Instead of following fixed routines, learned how to create flexible therapy programs that adapt in real time. Using ROS2 action servers, could write scripts for rehab exercises that could be customized to each patient-just like a real physiotherapist would do. This is where theory met practice: robots like TIAGO became more than machines-they became helpful partners in care.

Also studied seven-segment motion curves, originally used in industrial robotics, and saw how they could be applied in healthcare. It's like teaching a robot to move like a luxury car instead of a jerky go-kart-starting and stopping gently, and moving steadily in between. By breaking the motion into phases, tried to made the robot more durable and its movements more natural.

These lessons changed how to see robots. They're not just machines doing tasks-they can be made to work with people, in a way that's safe, thoughtful, and even caring.

2.8 Neurorehabilitation Techniques for Disorders of the Hemispheres

Rehabilitation plans for people with hemispheric brain injuries focus on promoting neuroplasticity and functional recovery by encouraging motor relearning that is based on use and sensorimotor integration. Constraint-Induced Movement Therapy (CIMT), Repetitive Task Training (RTT), and Mirror Therapy with visual feedback techniques are some of the most common methods used.

To force the use of the impaired side, Constraint-Induced Movement Therapy limits the use of the unaffected limb. This helps the brain reorganize itself and overcome learned non-use [23]. This intensive, task-based approach has made a big difference in motor function, especially for people who have hemiparesis after a stroke. Repetitive Task Training adds to this by getting patients to do repetitive, goal-directed tasks. This strengthens motor pathways and helps recovery through neuroplastic changes. Mirror and visual feedback therapies give patients more sensory input by showing them how to move their unaffected limb or giving them virtual feedback. This can help them learn how to move better and make the motor cortex more excitable [23].

There are many benefits to adding robotic platforms to these traditional neurorehabilitation models. Robots can consistently and intensely deliver therapeutic exercises while also objectively measuring how well patients follow instructions, how well they move, and how their bodies respond in real time. Robotic systems can change the difficulty, length, and type of a task based on performance metrics and biometrics like muscle activation or heart rate variability. They do this through built-in sensors and adaptive algorithms. This flexibility allows for personalized therapy plans that get the best results for recovery and can get around some of the problems with manual therapy, like therapist fatigue or bias in subjective assessments.

2.9 Medical Robots with ROS and TIAGO PAL

As more and more medical robots use ROS-based architectures, versatile platforms have been created to help with rehabilitation and patient care. Among these, Pepper by SoftBank Robotics stands out because it can interact with people in a more advanced way. It uses advanced social cues and empathy modules to make users feel more emotionally connected to it. The Stretch robot from Hello Robot is another example. It provides mobile assistance with strong ROS integration, mostly for logistical and support roles in healthcare settings. At the same time, different rehabilitation exoskeletons use ROS-driven control frameworks to offer precise and adaptable therapies for walking and upper limb function. This shows how versatile and useful ROS can be in medical robotic applications [21].

The TIAGO Personal Assistant Lightweight (PAL) robot platform is a great example of how to combine different types of assistive technologies for use in rehabilitation settings. TIAGO has a 6-degree-of-freedom (DOF) robotic arm, RGB-D and LiDAR sensors for full environmental perception, and advanced speech recognition and synthesis modules. This makes it possible for people to interact with it in a complex way and communicate with it in a simple way in both clinical and home settings. The ROS2 middleware is the foundation of its architecture, which makes it a modular, scalable, and strong software framework. This design naturally supports compliance control, real-time safety monitoring, and fault tolerance, which are all important for deployment near patients who are at risk.

Early clinical uses of TIAGO have shown good results in cognitive training, getting patients involved, and rehabilitation exercises that are focused on tasks. The platform's one-of-a-kind ability to combine physical and social assistance makes it a leader in the field of next-generation therapeutic robotics.

These ROS-powered platforms show how autonomous control, sensor fusion, and human-centered design are coming together to make assistive technologies that can change to meet the needs of different patients.

2.10 Robotic Methods for Neuropsychological Disorders

Socially Assistive Robotics (SAR) is a new and promising way to treat a variety of neuropsychological disorders by offering structured, engaging, and adaptable therapeutic interventions. For people with Attention Deficit Hyperactivity Disorder (ADHD) and Autism Spectrum Disorder (ASD), SAR platforms make interactions very structured, which helps them pay attention for longer periods of time and engage with others through personalized and predictable behavioral cues that traditional therapies may not always be able to provide [17]. These robotic interventions create a controlled space where attention and social skills can be built up over time, which improves learning outcomes.

When someone has Post-Traumatic Stress Disorder (PTSD) or Traumatic Brain Injury (TBI), robot-assisted therapy often includes log-based adaptive exercise programs that help the brain reorganize its pathways and build up its strength. SAR systems can promote neuroplasticity and cognitive restoration in ways that are hard to achieve with traditional methods by constantly watching how patients respond and changing tasks as needed. Also, using gamified and fun interfaces makes patients more motivated and likely to stick with their rehabilitation, which are very important for it to work.

Robotic systems help keep cognitive functions going in people with neurodegenerative diseases like dementia by giving them regular routines and memory exercises that are meant to improve spatial awareness and recall. Researchers have found that regular, predictable interactions like this can slow cognitive decline and improve quality of life by giving people meaningful stimulation and social support.

Controlled studies have shown that SAR-driven interventions work better than traditional rehabilitation methods. They lead to big improvements in therapy adherence, cognitive gains, and emotional well-being [24]. These results show how SAR could change the way personalized neuropsychological care is delivered and make it easier for more people to get it.

2.11 Research Gap and Hypothesis

There has been a lot of progress in the development of robotic neurorehabilitation systems [15], but there are still a number of problems that make them less useful in clinical settings. One big problem is that motor and cognitive rehabilitation modules are often designed and run as separate systems. This makes it impossible to take a truly integrated approach that deals with the many different aspects of hemispheric brain injuries. Also, current treatment plans are often rigid and standardized, so they can't change to meet the needs and progress of each patient. This rigidity can make patients less interested and make it harder to personalize treatment, which is necessary for successful rehabilitation. Also, patient engagement is often measured using broad metrics like task success or completion rates, without taking into account ongoing, quantitative measures of motivation, attention, or emotional involvement, which are important factors that affect how well therapy works.

To fix these problems, our socially assistive robotics (SAR) approach adds an integrated framework that includes hemisphere-specific tasks that are designed to help each patient's specific cognitive and motor problems. This system uses multiple types of patient inputs, such as gaze tracking, speech recognition, and motion capture, to change the difficulty of tasks and therapy settings in real time. This makes for a more personalized and adaptable rehabilitation experience. The framework includes important quantitative engagement tracking tools that look at more than just task performance to get a full picture of a patient's emotional state and responsiveness.

This flexible, multimodal approach will make a big difference in the results of rehabilitation. Specifically, expected that motor function recovery will improve by at least a few percent compared to traditional methods. Also expect that personalized and socially engaging therapeutic interactions will make patients more motivated and likely to follow through with their treatment. Also, using objective, ongoing clinical metrics will make it easier to consistently and accurately evaluate how well therapy works, which will make it easier to keep improving treatment protocols.

The next chapters go into more detail about how to use the TIAGO platform to implement this SAR framework based on ROS2. They also show experimental results that show how the system could change neurorehabilitation practices by offering personalized, responsive, and engaging therapeutic interventions.

Chapter 3

The TIAGO Robot

"I believe that inside every tool is a hammer."

Adam Savage

This chapter gives the information to communicate and interact with the PAL Robotics TIAGO robot using both terminal-based and web-based interfaces. It gives a guidline to connect to the robot's PC over a local Wi-Fi network, access it using SSH, and check the connection to make sure it works for remote operation. Moreover, the PAL Web Interface's functionality, like navigation, localization, battery monitoring, and location tracking. It also provides the tools and features that the robot's visual programming interface offers, which allows control movement, interaction, and behavior using a block-based programming method. These skills are very important for robotic applications to well, especially in research and assistive robotics settings.

3.1 Connecting to the Robot's PC

To remotely control the robot's PC from another computer, follow the steps described in the remainder of this section.

Connecting to the Robot's Wi-Fi Network

First, connect the PC/ Ipad/ Smartphone to the robot's Wi-Fi network. The robot's Wi-Fi network is typically named after the robot's ID; for example, `tiago-229c`. Make sure the PC is connected to this network, as this connection allows user to communicate directly with the robot's PC, So, that user can also access the Web Programing and Web Commander interface with any device like smartphone, Ipad, Laptops using the websites below:

```
http://tiago-229c:8080/
http://tiago-229c/index
```

Accessing the Robot's PC via SSH

Once connected to the robot's Wi-Fi, open a terminal on users PC. User will use the terminal to remotely access the robot's PC. To initiate a secure connection, use the SSH (Secure Shell) protocol, which enables user to control the robot's PC over the network securely. Enter the following command in the terminal as shown in the Figure 3.1:

```
ssh pal@tiago-229c
Password: pal
```

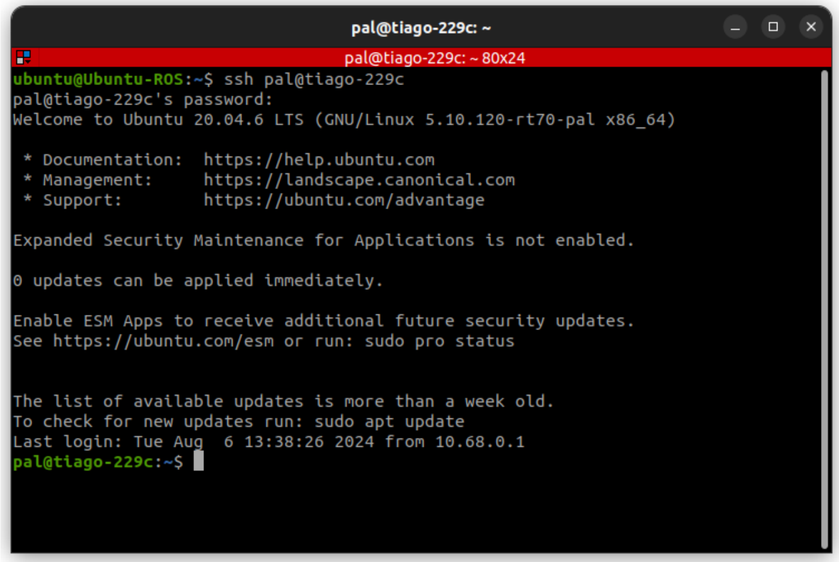


Figure 3.1: Accessing the PC of Robot Via SSH [1]

This will log user into the robot's environment, and user will see the terminal prompt change to `pal@tiago-229c:`. Now, user can perform tests or any necessary operations on the robot through the terminal.

For example, user can check the connection by using the command `ping tiago-229c`.

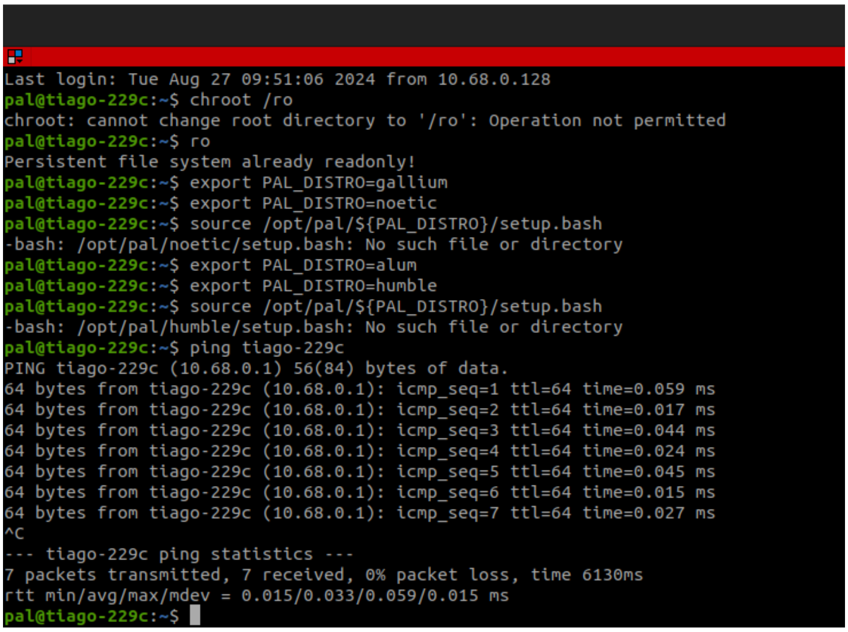
Controlling the Robot's PC Remotely

With the SSH connection established, the user have full access to the robot's PC. User can now perform various tasks, such as running diagnostic tests, updating software, or monitoring system performance. The remote connection allows user to control the robot's movements, monitor its sensors, and adjust its settings from users PC.

Verifying the Connection

To verify that users connection to the robot's PC is active, using network commands like `ping` user can check the communication status. For example, user can type the following command in the terminal as shown in the Figure 3.2:

```
ping tiago-229c
```



```
Last login: Tue Aug 27 09:51:06 2024 from 10.68.0.128
pal@tiago-229c:~$ chroot /ro
chroot: cannot change root directory to '/ro': Operation not permitted
pal@tiago-229c:~$ ro
Persistent file system already readonly!
pal@tiago-229c:~$ export PAL_DISTRO=gallium
pal@tiago-229c:~$ export PAL_DISTRO=noetic
pal@tiago-229c:~$ source /opt/pal/${PAL_DISTRO}/setup.bash
-bash: /opt/pal/noetic/setup.bash: No such file or directory
pal@tiago-229c:~$ export PAL_DISTRO=alum
pal@tiago-229c:~$ export PAL_DISTRO=humble
pal@tiago-229c:~$ source /opt/pal/${PAL_DISTRO}/setup.bash
-bash: /opt/pal/humble/setup.bash: No such file or directory
pal@tiago-229c:~$ ping tiago-229c
PING tiago-229c (10.68.0.1) 56(84) bytes of data.
64 bytes from tiago-229c (10.68.0.1): icmp_seq=1 ttl=64 time=0.059 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=2 ttl=64 time=0.017 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=3 ttl=64 time=0.044 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=4 ttl=64 time=0.024 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=5 ttl=64 time=0.045 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=6 ttl=64 time=0.015 ms
64 bytes from tiago-229c (10.68.0.1): icmp_seq=7 ttl=64 time=0.027 ms
^C
--- tiago-229c ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6130ms
rtt min/avg/max/mdev = 0.015/0.033/0.059/0.015 ms
pal@tiago-229c:~$
```

Figure 3.2: SSH Connection Check [1]

This command sends test packets to the robot's PC, and if the connection is working correctly, user will receive responses back. This feedback confirms that users PC can communicate with the robot's PC over the network.

3.2 Overview of the TIAGO Web Programming Interface

The PAL TIAGO robot is a versatile robotic platform designed for research, development, and various service based applications. Its programming interface allows developers to control and interact with the robot using a variety of software tools and languages. Here's an overview of the main aspects of the PAL TIAGO Robot Programming Interface.

Home Page Features of the PAL Web Interface

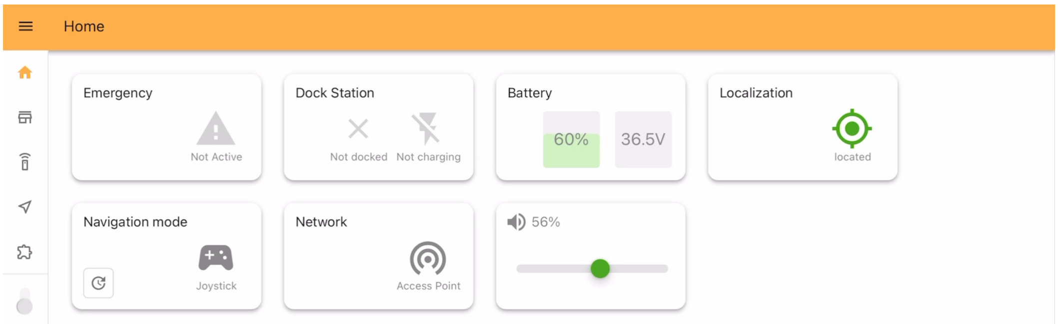


Figure 3.3: Home Page of Web Programming Interface [20]

The home page of the PAL web interface includes the following features as shown in the Figure 3.3:

- **Emergency:** This section displays the current status of the robot, indicating whether the emergency switch is on or off. It also provides alerts for any serious warnings related to the robot's hardware or connected devices.
- **Dock Station:** This section shows the status of the robot in relation to its docking station, indicating whether it is properly connected or not. It also displays the charging status of the robot, including whether it is currently charging. The position of the dock is referenced against a pre-scanned location in the room or environment.
- **Battery:** Here, user can monitor the battery level, including the percentage remaining and the current voltage.
- **Localization:** This feature allows user to locate the robot within the known environment that was previously scanned.
- **Navigation Mode:** If a joystick is connected, its status will be displayed here. The joystick will automatically disconnect after a certain period of inactivity if no commands are sent from it.
- **Network:** This section shows the current network access point to which the robot is connected.

- **Volume:** This displays the current volume level of the robot.

Mapping Page Features of the PAL Web Interface

The home page of the PAL web interface includes the following features as shown in Figure 3.4:

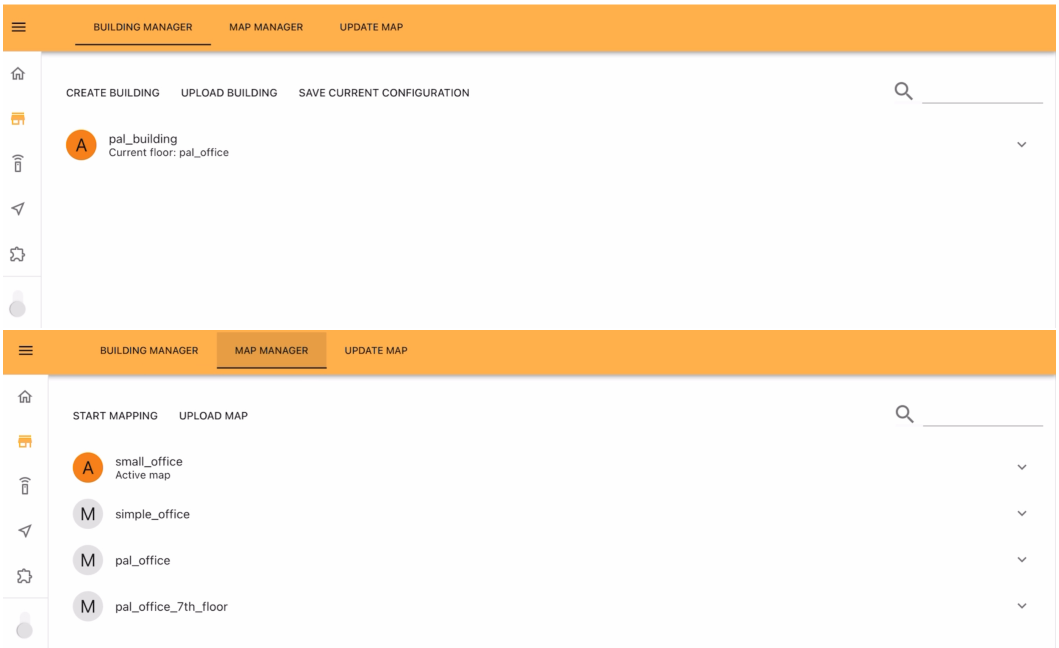


Figure 3.4: Mapping page of Web Programming Interface [20]

- **Building Manager:** This feature allows us to organize all the rooms or areas user have scanned under a single building or workspace. The robot will then know how to navigate from one room to another within this defined space. For example, as shown in the picture, for example a building named "pal_building." Using this user can create new buildings, upload existing building configurations, or save the current configuration.

- **Map Manager:** In this section, user can scan the working area or rooms where the robot will operate, creating a map for each room. These individual maps can then be combined in the Building Manager to display a complete map of all rooms on a single page. Here, user can start the mapping process using the proximity sensor and camera, or update a specific room's map to reflect any changes made.

- **Update Map:** This option allows us to update an existing map to accommodate changes required for the operations user need to perform.

Navigation WebGUI Page Features of the PAL Web Interface

The Navigation Web Plugin covers the user-robot interaction required for navigating the robot. It is composed of two different tabs as in Figure 3.5: Navigate, which allows interaction with the current navigation stack; and Edit map, which allows configuration of the map and its objects.

Figure 3.5: WebGUI Navigation Interface (Navigation Page) [20]

Throughout this chapter, the following terms will be used:

- **Point of Interest (POI):** A pose (position and orientation) in the map, identified by a name.
- **Point of Direction (POD):** An orientation with respect to the map, identified by a name.
- **POI List:** An ordered list of POIs and PODs.
- **Zone of Interest (ZOI):** An area in the map identified by a name.
- **Ramps (RA):** An area on the map which contains a ramp (change in ground level).

- **Virtual Obstacle (VO):** An area in the map marked as an obstacle for the robot, to prevent the robot from navigating in said area.
- **Highway (HW):** A preferred path for the robot, described as a polyline.
- **Draw on map:** An online editor that allows updating the map through drawing polygons and polylines directly in the map canvas.
- **Edit Map:** The Edit Navigate tab contains all the required tools to properly modify the robot's map in its known or unknown environment as shown in the Figure 3.6.

Figure 3.6: Defining Restricted Area using WebGUI Navigation [20]

The robot is shown by a pulsating orange arrow in the system's visual interface, allowing the identification of its current position. The laser scan data, which helps the robot in identifying nearby obstacles, is shown in pink. Points of Interest (POIs), shown by blue arrows, represent items or places relevant to task performance, whereas Points of Delivery (PODs), indicated by yellow arrows, indicate locations where the robot is expected to execute certain activities. Zones of Interest (ZOIs), representing certain areas that require the robot's focus, are shown by green polygons. Ramp regions, indicating accessible sloping surfaces, are marked with yellow polygons. Virtual Obstacles (VOs), representing impassable zones or transient obstacles, are marked by red polygons. Finally, the designated robot pathways, referred to as Highways, are depicted as black polylines, directing the robot along optimal routes.

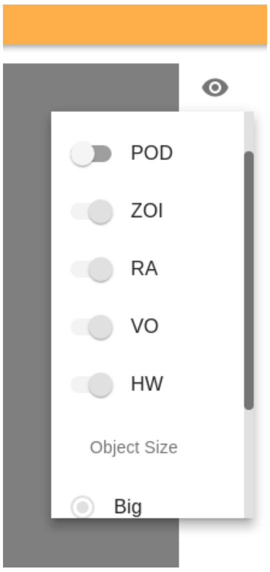


Figure 3.7: Mapping Features Using Navigation [20]

The map may be zoomed in and out by using the scroll wheel of the mouse. When the map is zoomed in, it can be moved by dragging it with the mouse wheel.

To display the name of an object (POI, POD, ZOI, RA, VO or Highway) hover the mouse above it for the label to appear Figure 3.7. If multiple objects are overlaid the name of each object will appear as well as its type.

3.3 Visual Programming Page Features of the PAL Web Interface

This page provides a clear overview of the visual programming features available in the PAL Web Interface. One of the main components is the "Blocks" section, which is organized into four categories: Actions, Conditionals, Control, and Decorators each offering specific functionalities for programming the robot's behavior. The New Program section allows users to drag and drop the desired blocks to visually design the flow of the program. The Variables feature enables the creation of custom variables, which can be used to communicate between different scripts or within the internal logic. Additionally, Constants are available and are particularly useful for motion-related activities, providing fixed values that help structure and control robot movements effectively shown Figure 3.8.

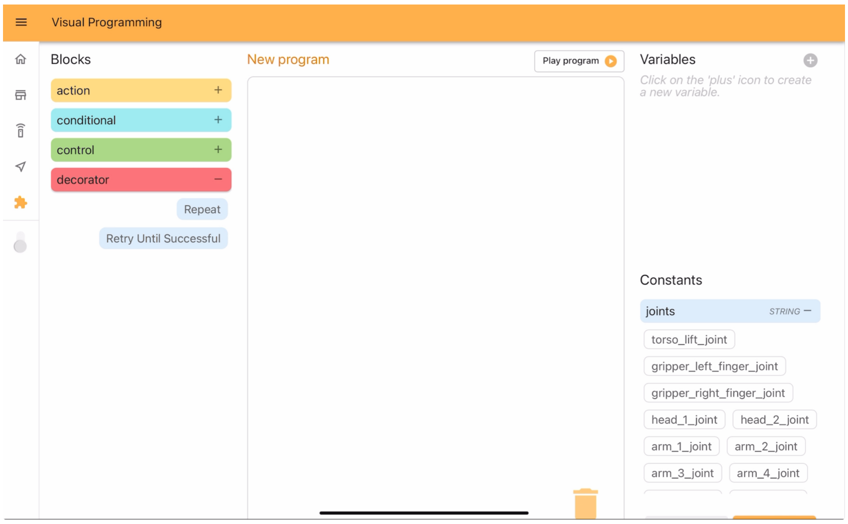


Figure 3.8: Visual Web Programming Interface [20]

3.3.1 Actions

Actions are executed by the robot and can either Succeed or Fail, The tools which can use in this section are discribe below Figure 3.9.

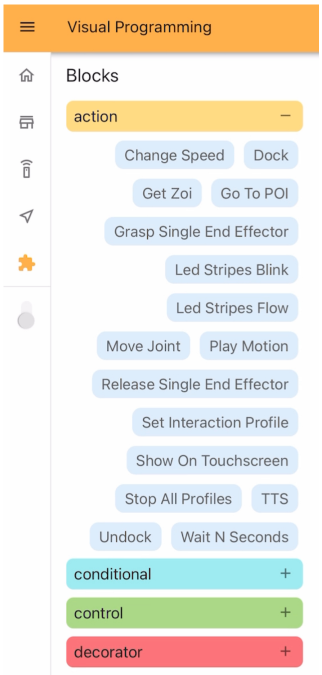


Figure 3.9: Action Block in Visual Web Programming Interface [20]

Change Speed: To change the velocity of the specifice motion such as arms, joints or wheels.

Dock: The dock action makes the robot connect to a dock station in front of it.

Get Zoi: Zone of Interest an area in the map identified by a name.

Go To POI: The Go To POI action instructs the robot to navigate to a POI in the current map. The action will SUCCEED when the robot reaches the target POI, if for any reason the robot is unable to reach the POI the action will FAIL.

Instruction	Meaning
target poi [String Input Port]	Name of the POI to navigate to.

Grasp single End Effector: Like a Flag to understand the Grasping status.

LED Stripes Blink: The Led Stripes Blink action instructs the robot to blink the leds in the configured pattern for some time, switching back and forth between two colors.

Instruction	Meaning
effect duration [float Input Port]	Total duration of the effect
first color [string Input Port]	Color in format R,G,B,A 255 (i.e. white is 255,255,255,255)
first color duration [float Input Port]	Duration of the first color
second color [string Input Port]	Color in format R,G,B,A 255 (i.e. white is 255,255,255,255)
second color duration [float Input Port]	Duration of the second color

LED Stripes Flow: To customize the LED flow on the Base of the robot.

Move Joint: To trigger a specifice motion of joint.

Play Motion: To trigger an complete motion of designed operation.

Release single End Effector: Like a Flag to understand the Grasping status.

Show On Touchscreen: The "Show On Touchscreen", action shows a list of buttons on the small TFT screen on the back of the TIAGO base, and then waits for a user to press one of the buttons.

Instruction	Meaning
options [string Input Port]	Comma-separated list of buttons to show on the TFT.
selected option [string Output Port]	The option selected by the user will be stored in this port.

Stop All Profiles: Like Emergency, To stop all runing scripts from web designer.

TTS: The TTS action will make the robot speak a sentence.

Instruction	Meaning
text [string Input Port]	Text to be read by the robot.
lang [string Input Port]	Language code in which the robot should speak (i.e. en GB for british english).

Undock: The Undock action will disconnect the robot from a dock station.

Wait N Seconds: The Wait N Seconds action will halt the execution of the program for the configured amount of time.

Instruction	Meaning
time [unsigned int Input Port]	Number of seconds the robot should wait.

3.3.2 Conditional

Conditionals are used to redirect the program flow according to different conditions. Conditionals must be used in chains that start with an IF, have zero or more ELSE IFs and may finish with an ELSE block shown in Figure 3.10.

Else: If the values of both ports match execute the children of this block.

Instruction	Meaning
value A [any Input Port]	A value to compare, it may be of any type, but both ports must have the same type.
value B [any Input Port]	B value to compare, it may be of any type, but both ports must have the same type.

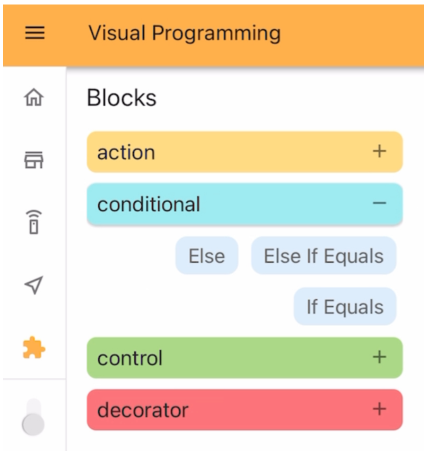


Figure 3.10: Conditional Block of Visual Web Programming Interface [20]

Else If Equals: If the values of both ports match execute the children of this block.

Instruction	Meaning
value A [any Input Port]	A value to compare, it may be of any type, but both ports must have the same type.
value B [any Input Port]	B value to compare, it may be of any type, but both ports must have the same type.

If Equals: In case all the previous IF and ELSE IF blocks in the chain have failed, execute this blocks children.

3.3.3 Control

Controls modify the behavior of the program sequence shown in Figure 3.11.

Fallback: Children of a Fallback block will be executed in order until one of them SUCCEEDS, and then it will SUCCEED. If no block SUCCEEDS then it will FAIL.

Sequence: A Sequence block encapsulates a set of blocks, executing them in order until one of them FAILS, in which case it will FAIL. If all of them SUCCEED it will SUCCEED as well.

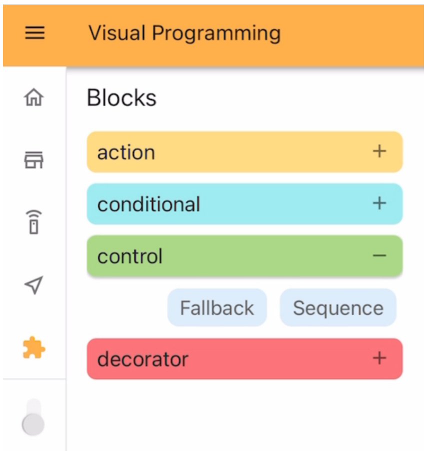


Figure 3.11: Control Block of Visual Web Programming Interface [20]

3.3.4 Decorator

Decorators modify how blocks behave on success and failure situations shown in Figure 3.12.

Repeat: A Repeat block will repeat its children blocks in order a number of times, as long as they SUCCEED.

Instruction	Meaning
num cycles [integer Input Port]	Number of times to repeat the children blocks.

Retry Until Successful: A Retry Until Successful block will keep executing its children actions in order until all of them SUCCEED, up to the configured number of times.

Instruction	Meaning
num attempts [integer Input Port]	Maximum number of tries, can be set to 0 to retry indefinitely.

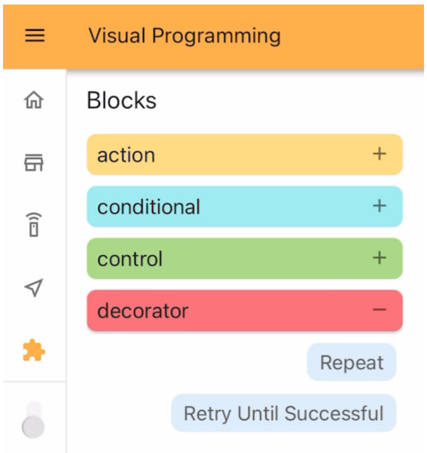


Figure 3.12: Decorator Block of Visual Web Programming Interface [20]

3.3.5 Variable Creation

The variable creation feature allows users to define and manage custom variables within the visual programming environment. These variables make the programming more flexible by storing values that can change during execution. Users can reuse the same variable across different programs, which helps keep the code organized and reduces repetition. This also supports better communication between different parts of the program or between separate scripts, making the robot's behavior more dynamic and easier to manage shown in Figure 3.13.

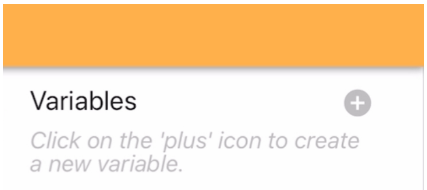


Figure 3.13: Variables Section to Define New Using Visual Web Programming Interface [20]

3.3.6 Constants

The constants feature provides predefined values that are especially useful for controlling the movement of the robot's joints and end-effectors. These constant values help ensure that the robot performs actions consistently and accurately. By using constants, users can avoid manual input of repetitive values and simplify the programming process, especially for motion-related tasks shown in Figure 3.14.

Figure 3.14: Constants Block Visual Web Programming Interface [20]

Chapter 4

Implementation

"Sometimes, the elegant implementation is just a function. Not a method. Not a class. Not a framework. Just a function."

John Carmack

This chapter explains how the PAL TIAGO robot was set up to perform rehabilitation tasks using both software and hardware components. Instead of using a visual, web-based programming system, the robot is controlled through a full ROS2-based setup built from scratch using Python. This system allows the robot to talk to patients, listen to voice commands, move its arms and base, recognize faces and gestures, and save session data for later analysis.

The design of this system is based on three important goals: **real-time responsiveness**, **modularity**, and **clinical safety**. Real-time responsiveness means the robot can quickly react to different types of input like voice commands, facial expressions, or how a person is sitting without delays. Modularity means the system is built as a collection of small, separate parts (called nodes), each doing one specific job like face detection, voice control, movement, or saving data. These nodes communicate with each other using ROS2's publish and subscribe system. Because of this, it's easy to test, replace, or improve any part of the system without affecting the others.

All of these parts are controlled by a main node that acts like a manager. This main program keeps track of everything that's happening, decides which features to run based on voice commands or test settings, and makes sure the system is running smoothly. It also handles switching between different modes, like setup, therapy, and shutdown.

Finally, because the robot works directly with people, safety is very important. Along with a physical emergency stop button, Have added a "panic word" feature in the voice system. If a critical word is spoken (like "stop" or "emergency"), the robot will immediately stop all actions and freeze in place to avoid any risk to the patient.

This setup creates a flexible, safe, and interactive environment that allows the TIAGO robot to help with therapy in a smart and semi-autonomous way. The rest of the chapter explains how each part was built and how they work together as one complete system anf for understanding better refer Figure 4.1.

4.1 System Overview

The system was designed for structured indoor environments such as clinics or hospital or rehabilitation centers. It facilitates both cognitive and physical therapy through the following capabilities for the system flow understanding refer Figure 4.1.

1. Multi-modal interaction (voice and web interface)

2. Visual perception using face detection and recognition

3. Arm trajectory execution with 16 therapeutic positions

4. Autonomous navigation between therapy zones

5. Performance monitoring and data visualization

6. Emergency stop and fault recovery for patient safety

The system operates through a finite state machine (FSM) implemented using ROS2 nodes, each responsible for a core function and synchronized using a multi-threaded executor.

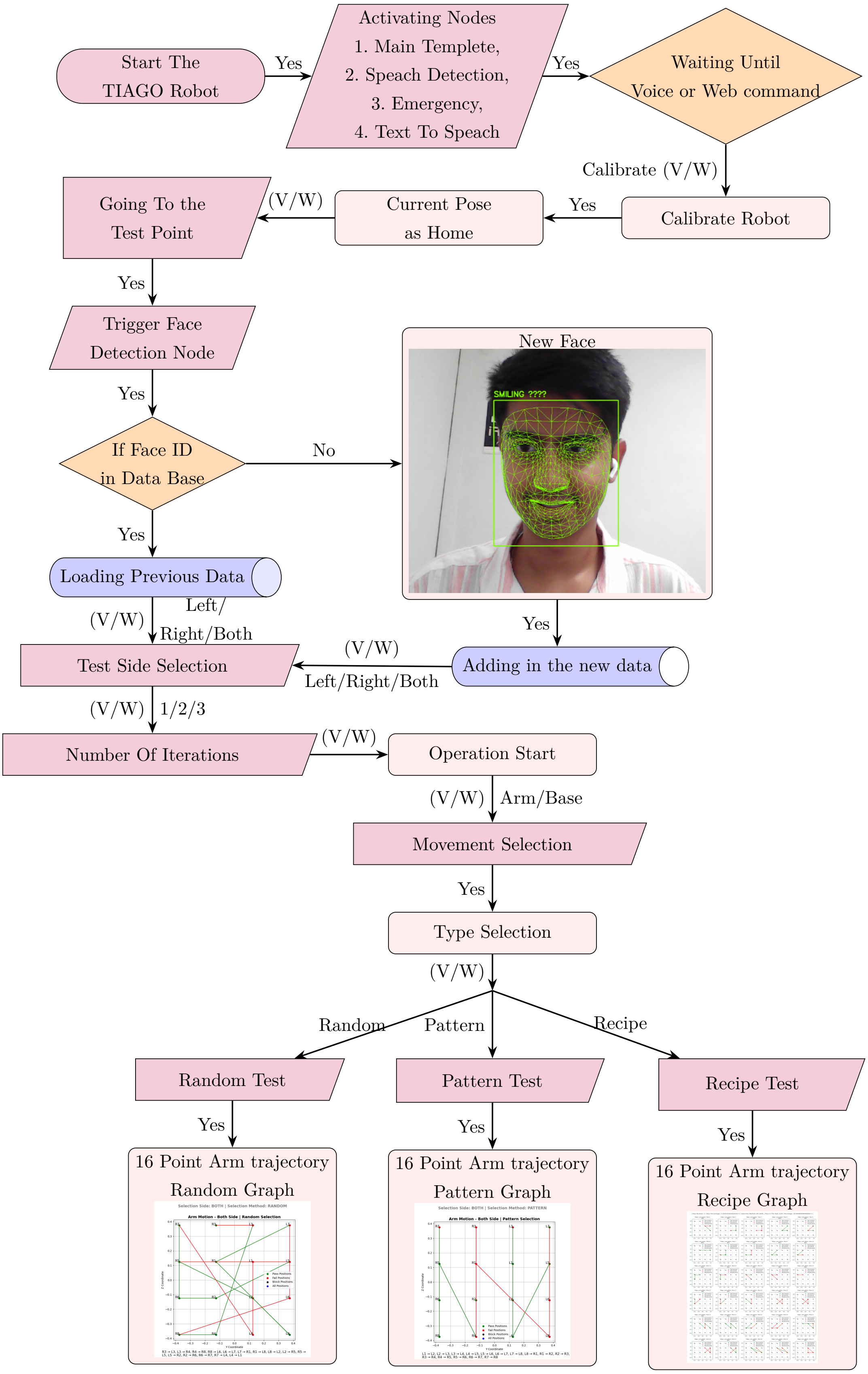


Figure 4.1: Execution Flow

4.2 Development Environment

The robot system used in this thesis is built on hardware and a modular software stack, for real-time responses and adaptability for rehabilitation tasks. Table 4.1 outlines the key components of the TIAGO robot, including its onboard computing power, sensing devices, actuators, and audio interfaces.

Table 4.1: TIAGO Hardware Specifications

Component	**Specification**
Onboard PC	Intel i7 CPU, 16GB RAM
Sensors	RGB camera, 2D LiDAR for environment perception
Actuators	6-DOF robotic arm (3kg payload capacity), mobile base for navigation
Audio	Microphone array for voice input, onboard speakers for system responses

The software stack includes all the important tools and systems needed to run and control the robot. Table 4.2 shows the main components used, such as the operating system, ROS2 middleware, simulation programs, and useful programming libraries.

Table 4.2: Software Stack Used in the System

Component	**Details**
Operating System	Ubuntu 22.04 LTS
Middleware	ROS2 Humble Hawksbill
Simulation Tools	Gazebo for 3D simulation, RViz2 for visualization
Programming Language	Python 3.11
Libraries	OpenCV (image processing), NumPy (numerical computation), SQLite (data storage), Vosk (automatic speech recognition), pyttsx3 (text-to-speech), matplotlib (data visualization), pyexcel_ods (data export)

4.3 Architecture and Execution Flow

There are various sections of the architecture that operate together when anything happens. Nodes can talk to one other through ROS2 topics, actions, and services. A `MultiThreadedExecutor` takes care of several threads at once. When user switch on the robot, it does the following:

1. To connect to the PAL Robotics online interface, open a browser and go to `https://localhost:8080`.
2. To get to the building map interface, user need to log in with the credentials user were given.
3. Start scanning the environment with the web UI.
4. To get 2D LiDAR data for SLAM mapping, user need to turn the robot around in the room at different positions in free space.
5. Find and mark by hand areas that are off-limits (such under tables or chairs) where LiDAR can mistake gaps for routes.
6. Break the room up into a virtual grid and give each point of reference a name (e.g., `RoomA_P1`, `RoomA_P2`) for navigation.
7. Save the map user made with a unique ID so user can use it again later.
8. To do robot localization, turn it in place until the current LiDAR scan matches a saved map.
9. Start the operational state with a self-localized position and identified reference points to help user find users way.
10. Keep doing face recognition, carrying out tasks, and logging data.

Listing 4.1: ROS2 Node Execution

```
executor = MultiThreadedExecutor()
executor.add_node(TextToSpeechNode())
executor.add_node(OperationControllerNode())
executor.spin()
```

4.4 Initialization and Calibration

The first thing user need to do to get the robot ready is to utilize a semi-manual mapping method to assist it learn where things are. This phase is especially crucial when user move the robot to a new site, such a hospital ward or a rehabilitation room. User only have to do it once for each new place.

1. **Web Interface Access:** To begin the mapping process, open the PAL Robotics online interface at `https://localhost:8080`. When users log in with the appropriate information, they see a number of setup options, including the space for making maps.

2. **Room Scanning:** In the "Building Map" section, user can turn the robot by hand to different parts of the room to obtain 2D LiDAR data from open spaces. This is the initial 2D map of the room made with SLAM (Simultaneous Localization and Mapping). Figure 4.2 shows the first map produced with the interface.

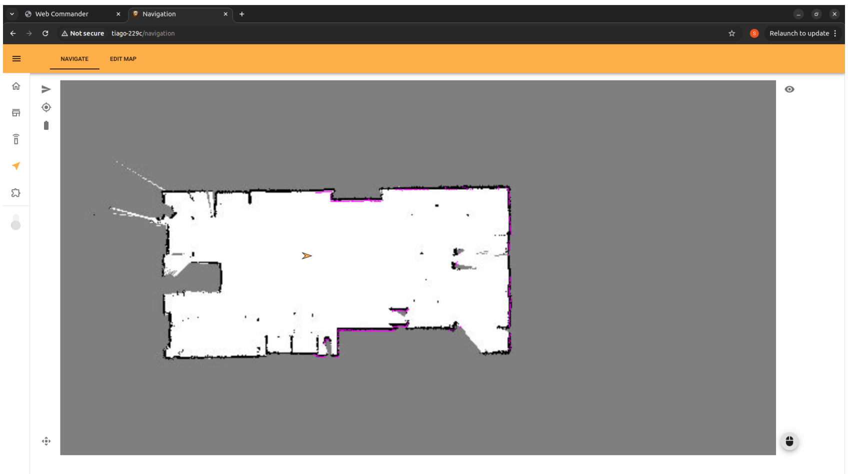


Figure 4.2: 2D Room Mapping Using LiDAR Sensor [20]

3. **Obstacle Refinement:** The 2D LiDAR sensor only scans things at a set height, which is usually close to the floor. Because of this, things like tables or chairs may look like they have clear passageways underneath them. Users can manually mark restricted areas on the map to avoid these kinds of misunderstandings. This helps the robot stay away from places that look like they are open but are really blocked. Figure 4.3 explains how to find restricted spaces after finding furniture like chairs.

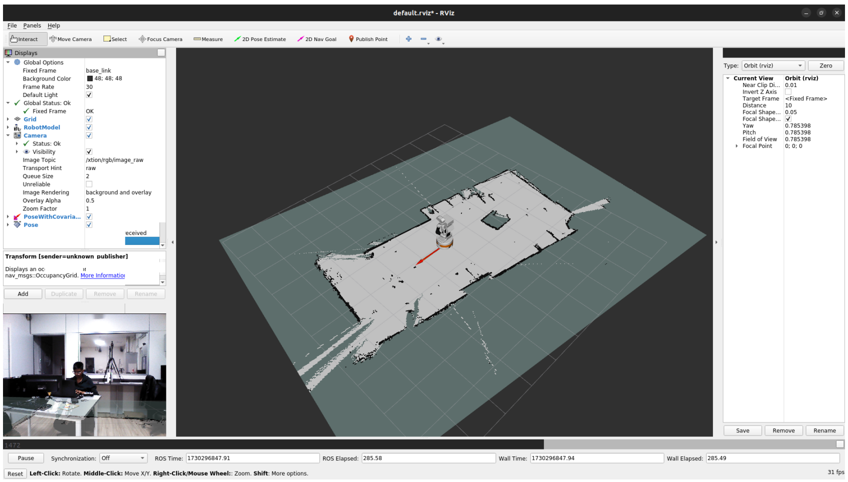


Figure 4.3: Manual Restriction of Obstacles Detected Incompletely by LiDAR [20]

4. **Point Definition:** In the mapped area, some reference points are set up to help with targeted navigation. The area is usually set up in a matrix or grid pattern, and each important place is given a name that makes sense, like `Room1`, `Room1_Bed2`, or `Door`. These named points let user give high-level commands like "Go to Room1_Bed2." Figure 4.4 shows the points on the map with green arrows showing which way the robot is facing at each point.

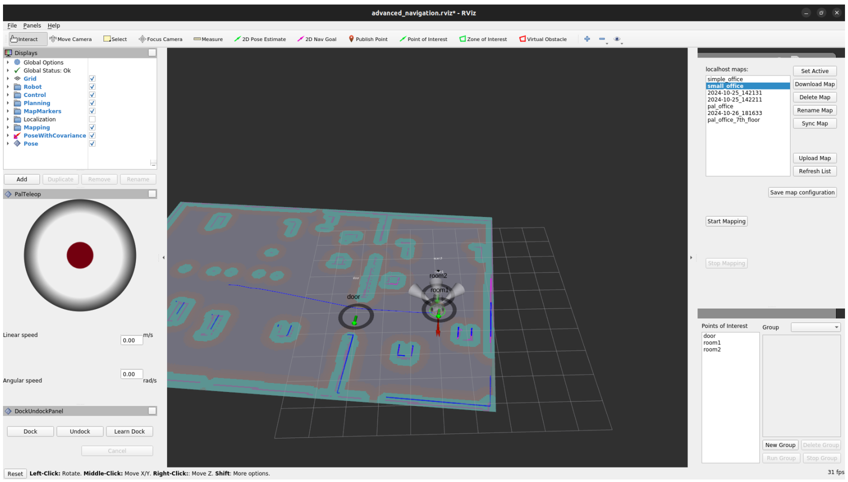


Figure 4.4: Definition of Navigation Points and Orientation in the Mapped Room [20]

5. **Map Saving:** After the scanning and settings are done, the map is stored with a unique ID. The interface usually asks users to save the map with information like the name of the building, the floor number, and the room ID.

This structure makes it easy to retrieve the map in the future and makes it possible to make small changes to it if needed. The map saving interface is shown in Figure 4.5.

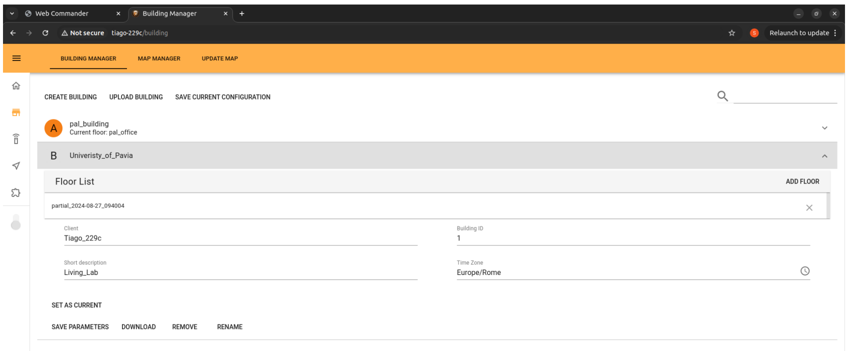


Figure 4.5: Saving the Mapped Environment with Descriptive Identifiers [20]

6. **Pose Calibration:** When the robot starts up, it rotates in position to find its own location. It compares the current LiDAR scan to the saved map to figure out where it is and how it is facing in the surroundings.

7. **One-Time Mapping:** User only need to do the complete mapping and point definition process once, and that's only for new environments. Only pose calibration is needed for future deployments in the same region, which takes a lot less time and effort.

This strong initialization process makes sure that the robot can navigate correctly, keeps it from pursuing unsafe or deceptive paths, and gives it a structured spatial layout for easy command creation. The system can adapt to changing settings and provide safe and dependable autonomous service by combining sensor data, human input, and a flexible map structure.

4.5 Voice and Web Command Interface

The robot system has an offline speech recognition pipeline that makes it easy to control with voice commands. This makes it easier to communicate with the robot in rehabilitation situations. This feature is especially useful for people who have trouble moving about or for places where direct touch is limited.

Offline Speech-to-Text using Vosk

The speech-to-text functionality is implemented using the open-source Vosk API, which supports offline Automatic Speech Recognition (ASR). This ensures that voice recognition remains robust and responsive even in environments with limited or no internet connectivity. The system leverages a small English-language model (vosk-model-small-en-us-0.15), which provides a balance between accuracy and computational efficiency, core Libraries used as follow.

- **vosk**: This is the offline ASR engine. It goes around Kaldi to support different languages and models.
- **pyaudio**: This library lets user transmit audio data from the microphone in real time so that it may be processed live. The ROS2 Python client libraries.
- **rclpy** and std_msgs.msg.String: ROS2 Python client libraries for publishing recognized speech as messages on a topic.
- **wave**, os, sys: Standard Python libraries for audio data handling, system operations, and file access.

Speech Node Structure

The node constantly records live audio and recognizes it in real time. Valid sentences are sent to a ROS2 topic called (/speech_to_text), where they are broken down into commands that can be used.

Listing 4.2: Speech Processing Callback

```
def listener_callback(self, msg):
    command = msg.data.lower()
    if "go␣to␣left␣point" in command:
        self.send_navigation_goal("left_therapy_zone")
        self.publish_tts("Navigating␣to␣left␣therapy␣zone")
```

The node that goes with this is shown below. It shows how streaming input is handled and turned into ROS messages

Listing 4.3: ROS2 Node for Speech Recognition

```
class SpeechToTextNode(Node):
    def __init__(self):
        ...
        self.stream = self.p.open(
            format=pyaudio.paInt16,
            channels=1,
            rate=16000,
```

```
        input=True,
        frames_per_buffer=8000
    )
    ...
    self.model = Model(MODEL_PATH)
    self.recognizer = KaldiRecognizer(self.model, 16000)
```

Command Recognition Pipeline

The command recognition pipeline begins by using the `PyAudio` library to continuously record the audio stream. These audio frames are streamed directly to the `KaldiRecognizer`, which processes the input in real time. Once a full phrase is detected, the system either gradually processes the recognized words, publishing them incrementally to a ROS2 topic, or it transmits the complete phrase at once. This ROS2 topic is concurrently monitored by a dedicated node that parses the incoming phrases to identify and extract predefined control commands, enabling responsive robot behavior based on voice input. his topic is also subscribed to by another ROS2 node, which then parses the content to find predefined control commands.

Advantages and Features

The speech interface offers several practical advantages that make it well-suited for use in clinical and therapeutic environments. One of its key features is offline operation, allowing it to function without requiring an internet connection an essential benefit in settings where connectivity may be limited or unreliable. The system is built with a modular architecture, where each component, from audio capture to command transmission, operates as an independent and replaceable block. This modularity not only enhances maintainability but also supports customization. Real-time responsiveness is achieved through continuous streaming and the ability to process partial recognition results, ensuring timely feedback during interactions. For safety, the interface allows users to define emergency keywords such as "stop" or "halt" which can trigger immediate safety protocols. Additionally, the use of open-source libraries provides both flexibility in development and cost-effectiveness. This voice interface works in tandem with a web-based graphical interface, offering users whether patients or therapists the choice to issue commands either through speech or by typing into a browser, thereby improving accessibility and user experience.

4.6 Camera, LiDAR and Motion Command Interface

The robot uses a combination of a camera, a LiDAR sensor, and motion control systems to understand its surroundings and move safely. The RGB-D camera captures both color images and depth information, which helps the robot recognize people, objects, and how a person is sitting or moving. This is important for interacting with patients in a more natural and personalized way. The LiDAR sensor scans the environment to create a map and detect obstacles, making it easier for the robot to move around without bumping into things. Both the camera and LiDAR data are processed through ROS2 nodes, which allow the robot to understand where it is and how to reach its destination. The motion command interface then takes commands either from voice input or a web interface and turns them into movement instructions for the robot's wheels. This system helps the robot move smoothly and safely, whether it's going to a specific location, avoiding obstacles, or approaching a patient when needed and perform the test as per the instruction.

4.6.1 Face Detection and Recognition

The face detection and recognition system enables the robot to interact meaningfully with individuals such as patients and clinicians. This is achieved using both standard and advanced ROS2-based methods. By using the predefined nodes from the `hri_body_detection` and `hri_face_detection` packages, the robot can estimate the distance between itself and the person, and assign every time new unique face ID, body ID, and person ID. These identifiers should compare with a pretrain model with the previous data to consistently recognizing individuals over time and understanding their posture or orientation as shown in the Figure 4.6 . The system is capable of tracking key facial landmarks even when some parts of the face are not clearly visible, thanks to robust, pre-trained models that can predict unseen regions.

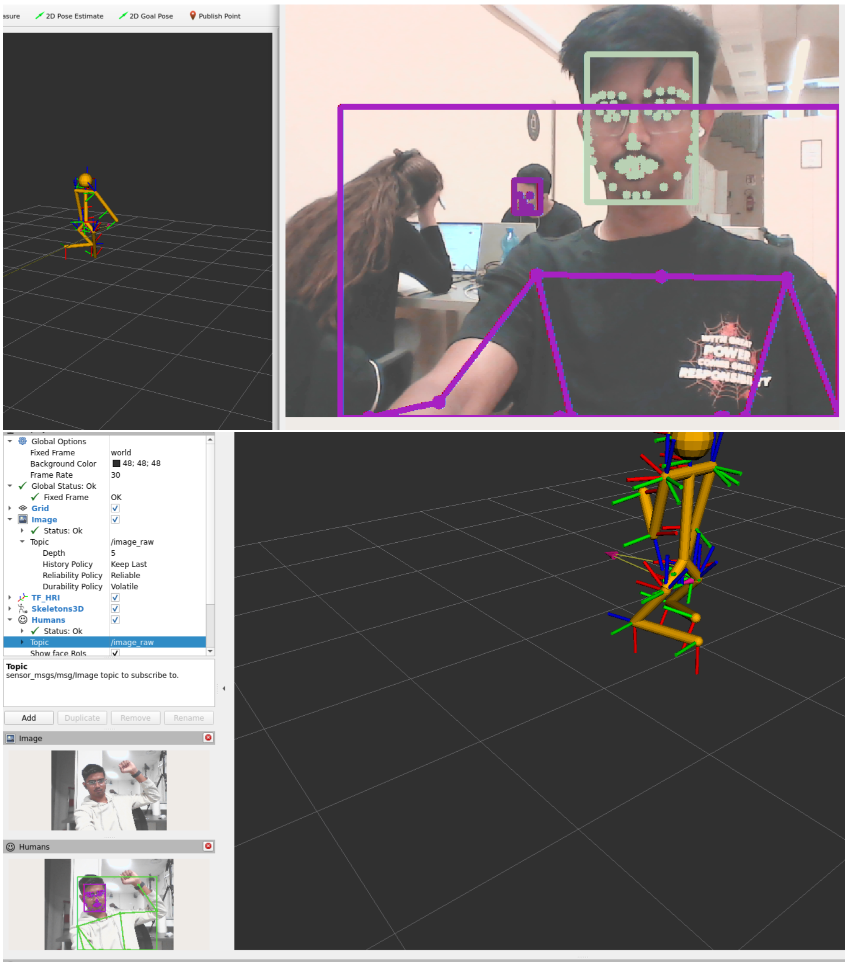
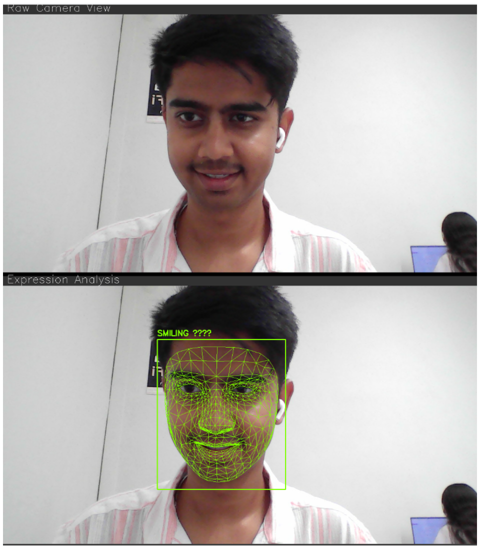
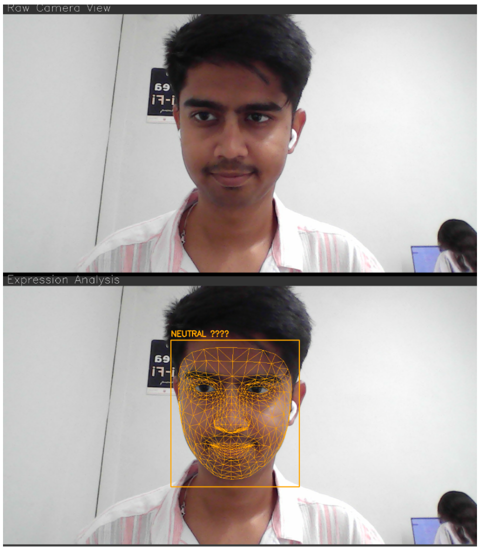


Figure 4.6: HRI Face and Body Dtection [1]

To enhance interaction during clinical tests, a custom module was developed to assess fatigue levels based on facial expressions as shown in Figure 4.7. This is done by analyzing the face through a mesh structure, allowing for a more detailed understanding of facial tension, focus, and attention during testing. Outside of testing periods, a default face recognition module is used to detect and label individuals. It uses either Haar Cascade classifiers or deep neural networks (DNNs) from OpenCV for face detection. This dual-mode system ensures efficient real-time face recognition, while also enabling deeper analysis when needed for patient assessment.

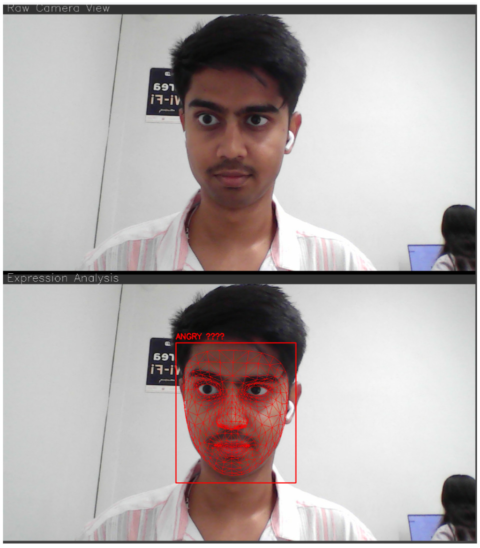


(a) Smiling Expression



(b) Neutral Expression

(c) Stressed Expression

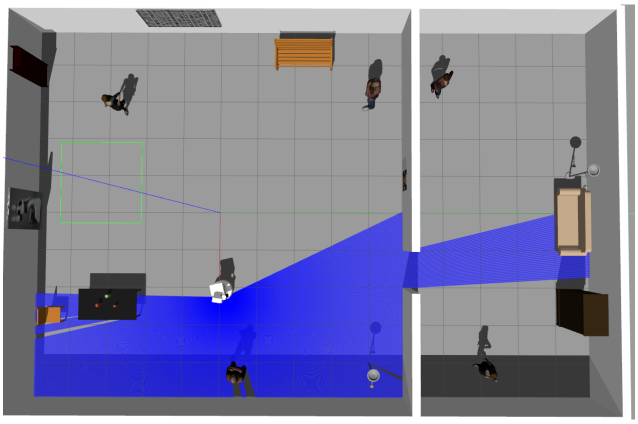


(d) Angry Expression

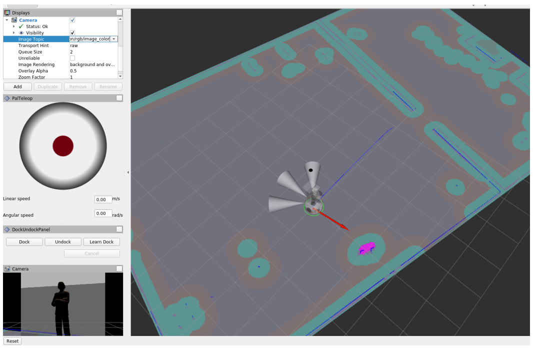
Figure 4.7: Facial Expression detection and recognition stages [1]

4.6.2 Navigation and Positioning

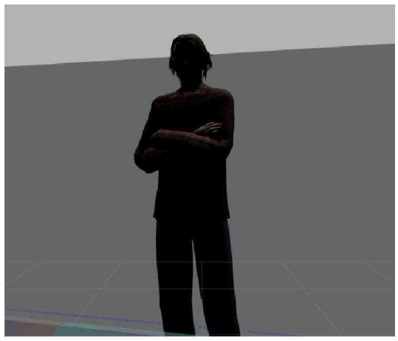
In this part, we explain how the robot understands its position and navigates around the patient. As shown in the top view image (Figure 4.8a), the robot is standing right in front of the patient. In the second image (Figure 4.8b), we can see how the robot sees the environment using its sensors. The human is shown in pink, and a light olive-colored area is drawn around them. This area acts as a safety boundary, so the robot does not move too close. Since the robot moves based on its center point and has a wide base, this space helps to avoid collisions.



(a) Top View



(b) Sensor Data



(c) Camera View

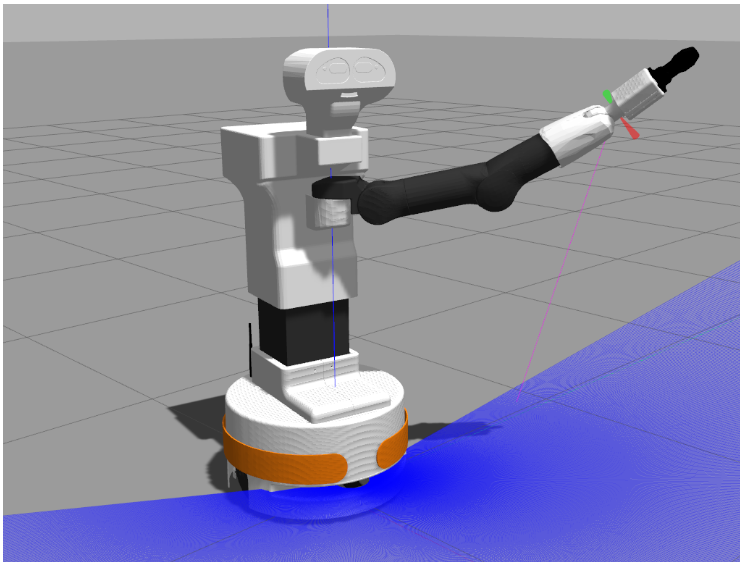
Figure 4.8: Navigation and Positioning [1]

We can also see the robot's path or trajectory where it came from and where it detected the human. Once the human is detected, their position is saved, and the robot adjusts its location accordingly. In the third image (Figure 4.8c), the robot's camera clearly shows the person in front of it.

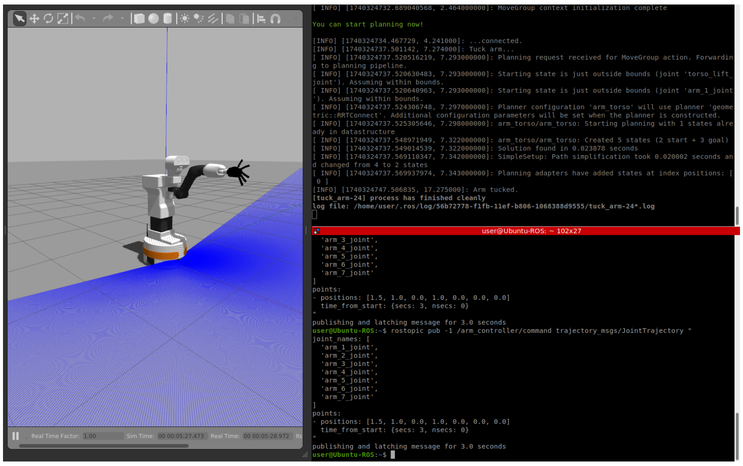
From the top view, we also notice the blue area that shows the LiDAR's field of view, which covers around 120-150 degrees. But when we combine data from the LiDAR, depth camera, and RGB camera, the robot's effective vision becomes narrower, around 70-90 degrees. This combined input helps the robot make better decisions and interact safely with the person in front of it.

4.6.3 Arm Trajectory Execution

To perform therapy tasks, the TIAGO robot uses predefined arm positions and moves through them with precise control. This is achieved using ROS2 action interfaces, where we control the robot's arm by sending joint angle commands for each specific pose Figure 4.9. Each arm (left and right) has 8 predefined positions L1-L8 for the left arm and R1-R8 for the right arm carefully selected to cover the required movement range.



(a) TIAGO Arm L7 Pos



(b) TIAGO Arm L3 Pos

Figure 4.9: Navigation and Positioning [1]

Each of these points represents a location the robot must reach during an exercise. The positions are arranged in a workspace map as shown Figure 4.10. This helps the system determine which sequence of positions to follow based on the therapy "recipe" designed for a patient.

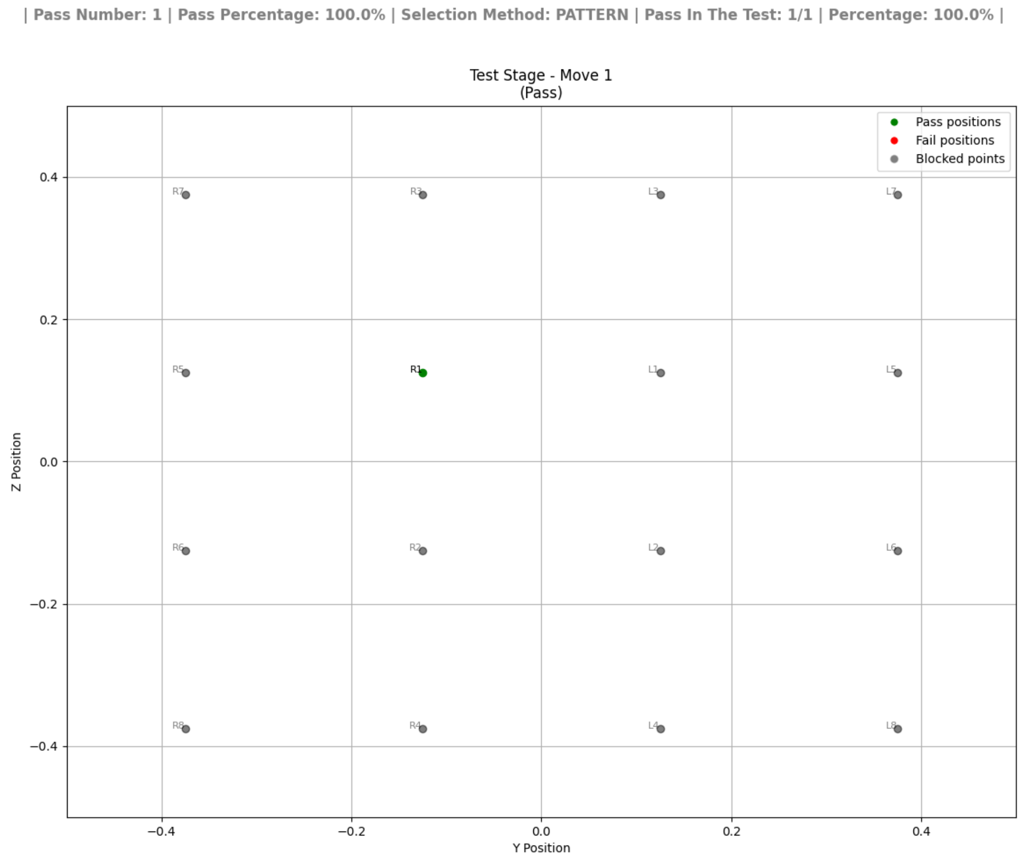


Figure 4.10: Therapy arm workspace with 16 predefined positions [1]

To execute these movements, we use a ROS2 node that sends joint trajectories to the robot's arm. The joint angles and speed (velocity) are defined in the message, and the robot moves smoothly between positions. Listing 4.4 is an example Python snippet that demonstrates how this movement is triggered.

Listing 4.4: Trajectory Generation

```python
def execute_random_movements(self):
    for pos in self.randomized_positions:
        msg = JointTrajectory()
        point = JointTrajectoryPoint()
        point.positions = [0.3, -0.125, 0.125, 0, 0, 0]
        msg.points.append(point)
        self.publisher.publish(msg)
```

This system allows the robot to carry out complex and repeatable movements with precision. The predefined joint targets ensure the arm stays within safe and functional limits during rehabilitation tasks. It also makes the setup highly adaptable for different types of therapies.

4.6.4 Recipes and Task Definitions

In this section, we have a predefined set of movements arranged in a 4 x 4 matrix. Each task level includes around 30 different arm movements. These movements can be simple, like moving from one point to another, or more complex, involving sequences of 3 or 4 positions. The level of difficulty increases based on the type of movement and how fast the robot is instructed to move.

All these tasks are defined in '.ods' files, which act like recipes. Each recipe contains the order and pattern in which the robot's arm should move. This makes it easy to create and adjust therapy sessions based on the patient's needs. The robot reads these files and follows the instructions accordingly, adjusting the motion and speed depending on the complexity of the task.

4.7 Data Management and Visualization

Effective data management and visualization are essential for analyzing the outcomes of robot-assisted therapy sessions. This section describes how data is stored, processed, and presented for both development and evaluation purposes.

Storing Data

The system collects various types of data during therapy sessions, including facial recognition logs, previous session outcomes, therapy duration, and position tracking information. These data points are downloaded and organized to support later analysis and system improvement.

Data storage is performed using two main strategies. First, structured information such as session logs and facial recognition results are stored using SQLite databases, allowing for efficient querying and relational organization. Second, visual and tabular data such as performance charts and session summaries are exported as ODS (OpenDocument Spreadsheet) files and image files (JPG or PNG) for external review and reporting.

Data Download: face recognition, Old results, therapy time, positional accuracy. Storage: SQLite for structured and recipes logs and ODS, JPG or PNG files for export are used for storage info.

Visualization

Visualization of performance metrics helps clinicians and researchers understand patient progress and system accuracy. The data collected is used to generate plots that represent the patient's movement patterns and engagement metrics. The following Python snippet illustrates how movement data is visualized using scatter plots. The resulting images are timestamped and stored for documentation.

Listing 4.5: Plotting Performance Data

```
def plot_movements(self):
    plt.scatter(y_vals, z_vals, color=result_colors)
    plt.savefig(f"Session_{datetime.now().strftime('%Y%m%d')}.png")
```

This visualization provides immediate feedback and is suitable for integration into progress reports shared with clinicians.

Therapy Metrics

To quantitatively assess the efficacy of each therapy session, a set of metrics is defined and monitored. These include the completion time of tasks, the spatial accuracy of patient movements, and the duration for which the patient's face remains in view indicating engagement.

Table 4.3: Therapy Metrics

Metric	**Measurement Technique**
Completion Time	ROS2 timestamps and logs
Movement Accuracy	Position error threshold ($<$ 5cm)
User Engagement	Face presence duration in frame

These metrics provide an objective basis for evaluating patient interaction and system performance over time (Not Completed in process).

4.8 Safety System

Safety is a critical component in any human-robot interaction scenario, particularly in therapeutic or assistive robotics. The system includes multiple layers of safety mechanisms to ensure user protection and smooth operation even in the presence of unexpected behavior or hardware faults. This section presents the implemented safety features, including emergency stopping and fault recovery procedures.

Emergency Stop

To protect users during therapy sessions, an emergency stop mechanism is implemented to immediately terminate all ongoing processes in response to voice commands. This allows both patients and clinicians to intervene at any moment should they perceive a risk.

The system continuously listens for a predefined set of keywords such as "stop", "accident", or "abort" shown in the Listing 4.6. When such a command is detected, all running processes associated with the current session are forcefully terminated, halting any robot activity.

The Python callback function responsible for this behavior is shown in Listing 4.6. It checks incoming voice messages and, if a matching keyword is found, invokes termination routines on all subprocesses.

Listing 4.6: Emergency Stop Callback

```
def emergency_stop_callback(self, msg):
    emergency_keywords = [
        "stop", "halt", "cease", "end", "terminate", "pause",
        "discontinue", "quit", "abort","emergency", "crisis",
        "urgent", "critical", "danger", "alert", "alarm",
        "contingency", "panic", "priority", "accident",
        "give_up", "relinquish", "forfeit", "desert", "abandon",
        "leave", "withdraw", "surrender", "vacate", "renounce",
        "scrap"
        ]
    if msg.data.lower() in emergency_keywords :
        for process in self.process_list:
            process.terminate()
```

This emergency response mechanism ensures immediate interruption and prioritizes user safety over task completion.

Fault Recovery

Despite robust design, system faults can occur due to hardware limitations or unexpected conditions. The fault recovery mechanism is designed to minimize disruption and restore safe operational status automatically or with minimal intervention. Specifically, the system includes the following recovery strategies:

- If it fails due to major problems, it will automatically go back to initial stage.
- Reset the fault depending on a timeout (30 seconds of inactivity) for arm movement or base movement after triggering the action.
- Hardware problems are watched over by a watchdog timer (Internal).

These features contribute to the system's robustness, ensuring that faults do not lead to hazardous situations or downtime.

4.9 Testing and Integration

Testing and integration are essential phases in ensuring the reliability and robustness of the robotic therapy system. A systematic approach was adopted to verify the functionality of individual components and their interactions within the integrated environment.

Strategy

The testing strategy used with isolating each software node and verifying its behavior using mock inputs to simulate real world signals. This unit testing phase helped identify issues early and reduced debugging complexity. Subsequently, integration testing was performed using ROS2 launch files, which allowed multiple nodes to be launched together in controlled scenarios. This approach facilitated the detection of communication errors and synchronization problems between nodes before deployment in physical environments.

Validation

Validation of the system proceeded through a two-stage process. Initially, simulation tests were conducted using Gazebo, providing a safe and repeatable environment to assess the robot's performance under varying conditions. Key parameters such as motion accuracy and face recognition alignment were evaluated within this virtual space. Following satisfactory simulation results, the system was deployed in a controlled laboratory setting for real-world validation. Adjustments to face recognition coordinates specifically the positioning of the mouth, eyes, and nose were fine tuned based on observations to enhance detection accuracy and system responsiveness.

Integration Challenges

Integrating different subsystems into one working system brought several challenges. One major issue was voice command recognition. The system sometimes responded slowly, so the Vosk speech model was preloaded to reduce delay. However, Vosk also had trouble understanding different English accents, which sometimes caused voice commands to be missed or misinterpreted. Another problem was the robot arm's

Table 4.4: System Challenges and Resolutions

Issue	**Solution**
Voice latency	Local model preload (Vosk)
Arm jitter	Waypoint interpolation
Face detection failures	Adaptive lighting threshold

jittery movements, which made its actions less smooth. The points need to provide to the robot are in radians, so must have to convert the x, y, z coordinates into joint angles for the 6-axis arm. This was solved by using waypoint interpolation to create more natural motion. Finally, changes in lighting affected face detection accuracy. This was improved by adding an adaptive threshold to help the system adjust to different lighting conditions. Table 4.4 shows a summary of these issues and how they were resolved.

4.10 Evaluation of Performance

To assess the effectiveness and reliability of the system, several key performance metrics were evaluated.

1. **Trajectory Accuracy:** The robotic arm was able to follow planned movement paths with high precision. During simulated testing sessions, it maintained accurate trajectories 90% of the time, demonstrating reliable control and positioning during therapeutic exercises on simulator.

2. **Voice Response Time:** The system's average response time to voice commands was measured at 0.5 to 1.5 seconds. This delay includes processing input through the Vosk speech recognition model and executing corresponding actions, and it falls within an acceptable range for interactive use.

3. **Safety Compliance:** Emergency stop commands were successfully detected and executed in all test cases. The system achieved 100% compliance with safety protocols, ensuring that critical voice commands like "stop" or "abort" triggered immediate stop in operation.

4.11 Summary

The PAL TIAGO robot has been successfully modified to perform semi-autonomous neurorehabilitation tasks within a controlled and interactive environment with Gazebo.

The system demonstrates real-time responsiveness to multimodal inputs such as voice commands, facial detection, and positional tracking, ensuring clinical feasibility. Its modular design allows easy debugging, flexible upgrades, and clear separation of functions such as movement control, user interaction, and data management. Patient safety has been prioritized through an effective emergency stop mechanism, ensuring the robot stop's all actions immediately upon detection of critical voice commands or abnormal behavior.

Currently, most of the planned features have been implemented and verified, particularly through simulation and partial deployment. Data is being saved using local databases now, which store therapy logs and metrics. Further the data will be saved in SQLite as reference to analyze performance trends or adjust future tasks. Additionally, graphical results and performance metrics are saved locally as image files for later analysis. While the core functionalities such as trajectory planning, voice interaction, and face tracking are operational, some tasks remain ongoing. One such feature is a web-based interface, currently under development, which aims to provide an alternative control method. This interface will allow users or clinicians to trigger commands directly via a browser instead of relying solely on voice input. Another ongoing effort is the full automation of recipe execution on the robot; although recipe parsing and execution mechanisms are partially functional, end-to-end integration is still in progress.

Overall, the project is in its final development phase of templete creation, with most components functioning as intended. The remaining integration work and user interface refinements are expected to be completed shortly, enabling a fully functional semi-autonomous therapy assistant.

Future Work

Building upon this foundation, several enhancements are planned. One key direction is the use of reinforcement learning to dynamically adapt the difficulty level of rehabilitation tasks based on user performance. Incorporating gaze and emotion tracking could further personalize interactions and improve the quality of human robot interaction (HRI). Finally, extending the system to support cloud-based dashboards would enable therapists to monitor patient progress remotely, compare metrics across sessions, and make data-driven decisions for therapy adjustments.

Chapter 5

Results

"Somewhere, something incredible is waiting to be known."

Carl Sagan

This chapter presents the key outcomes of the project, focusing on how well the system performed and how it handled different rehabilitation tasks. First, we discuss the system's performance metrics, such as how fast and accurately the robot responded in the simulated environment. Then, we explain how the robot executed rehabilitation tasks like base movement and arm motion using fixed scripts that simulated patient behavior. Following this, a comparative analysis is provided to evaluate the success rate across various modes of operation including random and patterned arm movements and circular base navigation. This analysis highlights how reliably the robot was able to reach its intended goals during different tests.An error analysis further explores small inaccuracies that occurred during task execution, such as slight drifts in movement or occasional misalignment, and reviews how well the voice recognition system responded to simulated commands. Finally, the chapter outlines several limitations of the system. These include delays compared to human therapists, limited adaptability to patient behavior, constraints in onboard data handling, and issues like sensitivity to lighting during face recognition or limited battery life. Together, these observations help provide a realistic view of what the current setup can achieve and what needs improvement for future real-world deployment.

5.1 System Performance Metrics

Table 5.1 shows how long it took for the robot to perform specific actions during movement, both in the simulated environment (using Gazebo) and in the real world with the actual robot. The purpose of this comparison is to understand how closely the simulation matches reality and to identify where delays or unexpected variations may occur.

Table 5.1: Execution time benchmarks

Operation	**Simulation**	**Real-World**
Arm Trajectory Execution	3.0s $\pm$0.5s	3.0s $\pm$2.0s
Base Circular Movement	3.5s $\pm$8.5s	3.5s $\pm$10.5s
Face Detection Latency	120ms	200ms

For arm trajectory execution, the robot took about 3 seconds in both simulation and reality. However, the time in the real world varied more ($\pm$2.0s compared to $\pm$0.5s in simulation). This variation is mostly due to small physical delays, motor calibration differences, or safety checks that are not fully represented in simulation.

In the case of the base circular movement, both environments showed an average of 3.5 seconds, but again, there was much greater variability in real-world performance. In simulation, it fluctuated by about 8.5 seconds (depends on the distance), while in the real world, the timing could swing as much as 10.5 seconds (depends on the distance). This happened because the robot sometimes had trouble with slippery surfaces, unexpected obstacles, or localization errors that don't exist in the controlled simulation.

For face detection latency, the robot detected faces faster in simulation (120ms) than in the real world (200ms). This is expected, as simulations run in ideal conditions without actual camera data processing or hardware delays. In real life, things like lighting changes, camera quality, and computation load on the hardware make face recognition slightly slower.

These differences show that while simulation is useful for developing and testing robotic behavior, real-world performance can vary due to physical and environmental factors. That's why it's important to test everything on the real robot before clinical use.

5.2 Rehabilitation Task Execution

The following sections describe the rehabilitation tasks implemented and tested using the TIAGO robot in a simulated environment. All actions were performed within the Gazebo simulator, and patient interactions were mimicked using fixed input scripts to represent different response behaviors. These simulations allowed us to test and evaluate how the robot would behave during real rehabilitation sessions, especially focusing on motion control, interaction logic, and response tracking. Although no real patients were involved, the system was designed to handle realistic scenarios, including gaze detection, contact success, and trajectory execution. The tests were split into three major parts: base movement navigation, arm motion across a fixed matrix, and recipe based arm movements with varying levels of difficulty. These simulated trials provided valuable insights into the system's robustness, repeatability, and how it could eventually be adapted for real-world therapy with human patients.

5.3 Base Movement Navigation

The robot is programmed to follow a set path that is shown in the image below. This path is meant to keep the robot in contact with the patient at all times during the test. The robot may move point by point in a controlled way along this path by using its neck and motorized base. The movement starts at a central point, which is defined as 0 degrees and is directly in front of the patient. It then moves to the left and right sides in little steps.

The robot moves in a symmetric arc that goes from -40 degrees (left) to +40 degrees (right), with each interval spaced 5 degrees apart. The robot starts at 0 degrees and moves 5 degrees to the left, then back to the center. Then it moves 5 degrees to the right and back to the center again. The left-center-right process goes on in 5-degree steps until the robot achieves the full arc range or loses contact with the patient.

Either a set number of iterations per point or a dynamic stop if the robot loses contact or the patient isn't paying attention control each test session. Also, every session always starts from the center (0 degrees) to make sure that everything is the same and can be repeated.

The system has a feature that lets user choose the maximum number of iterations at each test point to make the testing more reliable. This means that the robot will stay where it is until the count is met, even if it takes several tries to make or keep contact.

Using five main indicators, the graph for this exam shows a visual depiction of the results in Figure 5.1 to Figure 5.3:

- **Blue star:** Show the position of the patient.
- **Black dots:** Show the robot's base block positions during the test.
- **Red dots:** Show the planned trajectory points or target positions, connected with a line to show the intended path.
- **Green dots:** Show the positions where the contact was successful (pass points).
- **Yellow triangle:** Show the real-time position of the robot during testing.

The x-axis of the graph shows how far the robot is from the patient in meters, while the y-axis shows lateral positions, which show motions to the left, center, and right of the patient.

At the top of the graph, there is information regarding the test session, such as:

- **Selection Side:** Left, Right, or Both
- **Selection Method:** Pattern, Random, or Recipe
- **Movement Type:** Whether the test was performed using base motion or arm motion

This method of testing makes sure that it can be done again and is safe. It also lets user keep track of and analyze the patient's interactions and responses during rehabilitation exercises.

Base Movement Navigation with the Both-side test in Pattern

Figure 5.1 test checks to see if the robot can follow the whole trajectory pattern, which goes both left and right. It helps check how responsive the patient is across the entire arc range, making sure that the session is symmetrical.

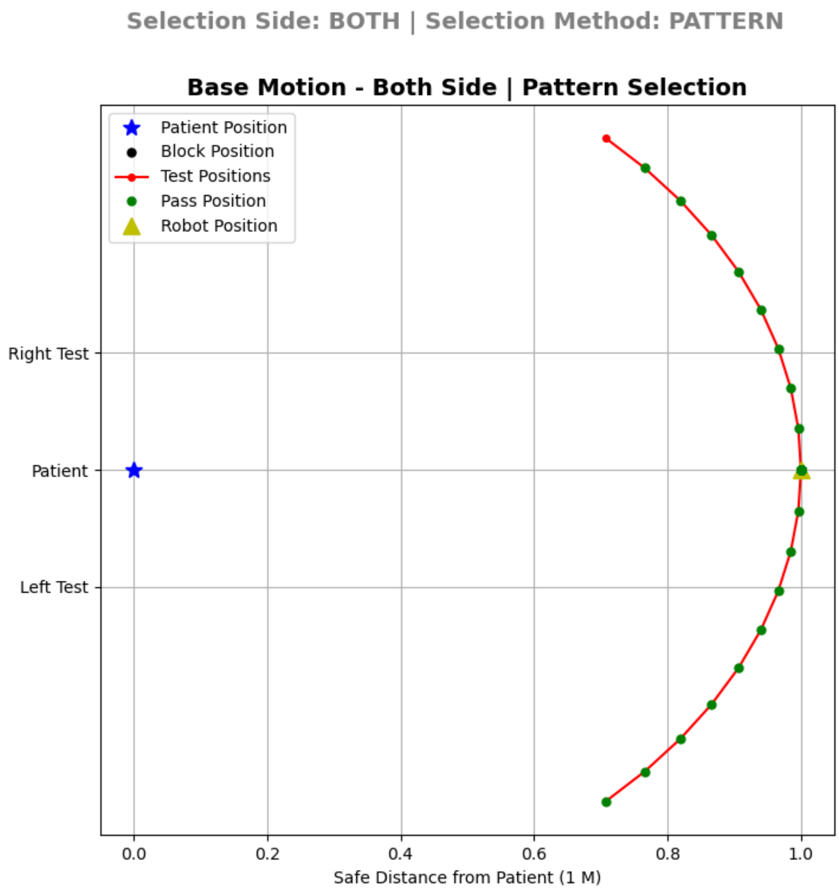


Figure 5.1: Circular navigation test showing 18 target points around the patient (radius=1m) [1]

Base Movement Navigation with the Left-side test in Pattern

Figure 5.2 test just looks at the left-side movement pattern, so it can focus on how the patient interacts with one side. It is helpful when mobility or engagement is predicted to be uneven or needs concentrated therapy.

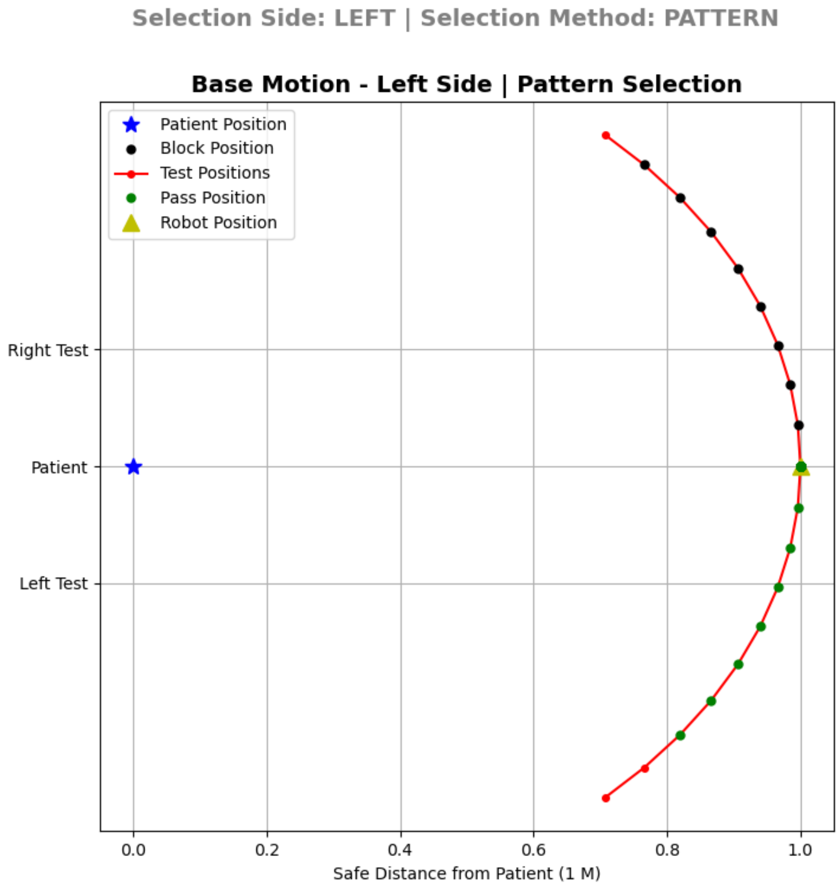


Figure 5.2: Circular navigation test showing 9 left side unblock target points around the patient (radius=1m) [1]

Base Movement Navigation with the Right-side test in Pattern

Figure 5.3 test only lets the trajectory go to the right side, which makes it easier to see how the patient reacts and stays consistent when they deal with stimuli or contact that starts on the right.

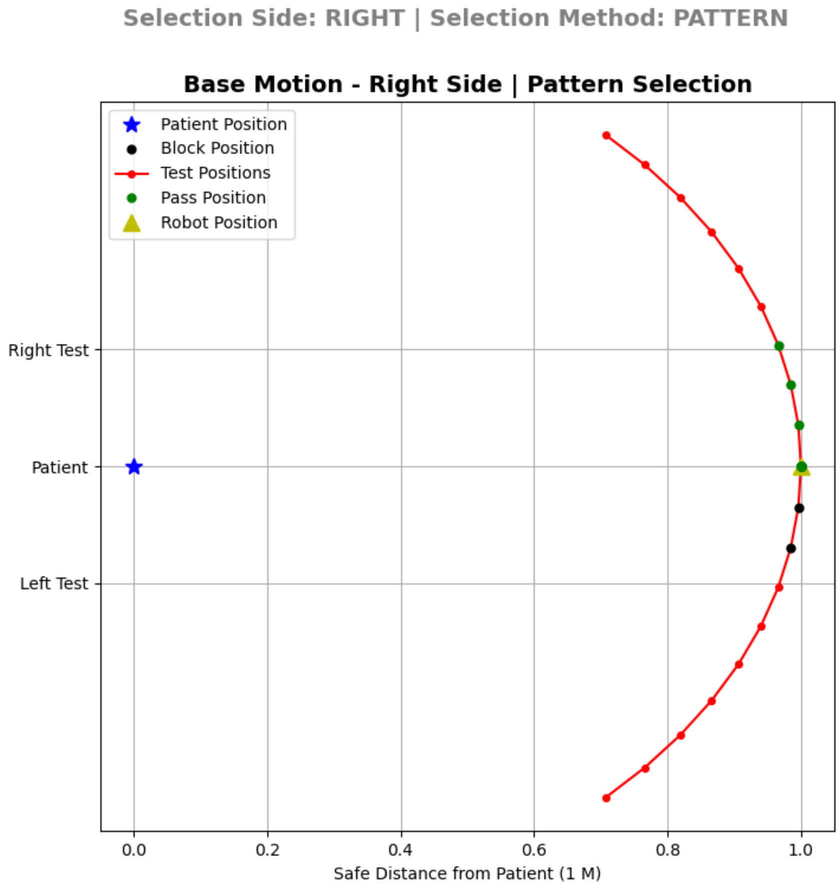


Figure 5.3: Circular navigation test showing 9 right side unblock target points around the patient (radius=1m) [1]

5.4 Arm Movement Patterns and Randoms

Have put up a test with a 4×4 grid (encompassing a 0.8m 0.8m × 0.8m 0.8m square area), which gives us 16 different spots for the robot to work in. These positions are split into two groups for targeted interaction:

- **Right Side:** First two columns of the matrix.
- **Left Side:** Last two columns of the matrix.

The arm movement challenge asks the robot to move its end-effector from one point to another in this matrix along a path that is either set or random. The robot always starts on the left and then moves to the right. Using a logical label-based structure, the point sequence is defined as follows:

$$\text{Working Area} = \begin{bmatrix} \text{R7} & \text{R3} & \text{L3} & \text{L7} \\ \text{R5} & \text{R1} & \text{L1} & \text{L5} \\ \text{R6} & \text{R2} & \text{L2} & \text{L6} \\ \text{R8} & \text{R4} & \text{L4} & \text{L8} \end{bmatrix}$$

To manage the testing procedures, two distinct movement strategies are implemented as described below:

- **Pattern-Based Movement** The robot proceeds through sympathetic through the matrix points in the pattern-based method. For example, it might go from L1 to L2 to L3, and then go from R1 to R2 to R3, and so on, until it finishes the selected side.
- **Random-Based Movement** When the system is in random selection mode, it uses Python's 'random' module to choose a group of target points. The user sets the number of points, and the algorithm makes sure that no point is used more than once until all of the selected points have been covered. This makes things more interesting and helps figure out how the patient reacts to unanticipated paths.

Test Configurations The system can run the test in six different ways: which are Both-side: Pattern-based, Both-side: Random-based, Left-side: Pattern-based, Left-side: Random-based, Right-side: Pattern-based, Right-side: Random-based can see the Figure 5.4 to Figure 5.9.

Tracking Contacts The robot's visual system keeps an eye on the patient at all times during each movement. To see if the patient is gazing at the robot's end-effector, their eyes and body are watched. If the two people make eye contact at the target location, the interaction is a pass. If not, it is seen as a failure.

Graphical Representation The graph that goes with the test shows the following indicators in a visual way from Figure 5.4 to Figure 5.9:

- **Blue dots:** Shows positions of the matrix to be visited (target array).
- **Black dots:** Shows robot's arm base positions during the test.
- **Red dots:** Shows failed trajectory points (where contact was not detected).
- **Green dots:** Show successful contact points (passes) and the corresponding trajectory path.

The Y-axis and Z-axis of the graph show the robotic arm's working area in meters, while the X-axis stays the same because the arm can't reach too far in front of it.

The following metadata is shown at the top of the graph:

- **Selection Side:** Left, Right, or Both
- **Selection Method:** Pattern, Random, or Recipe
- **Movement Type:** Arm Motion (for this section)

This way of testing makes sure that the results are very repeatable, safe, and easy to understand, which makes it a good way to measure how engaged patients are during upper-limb rehabilitation.

Arm Motion with the Both-side test in Pattern

In this test, the robot's single arm performs movements to all 16 predefined points 8 on the left and 8 on the right following a fixed sequence. This Figure 5.4 pattern-based motion ensures balanced interaction across both sides and helps assess the patient's symmetrical response to rehabilitation tasks.

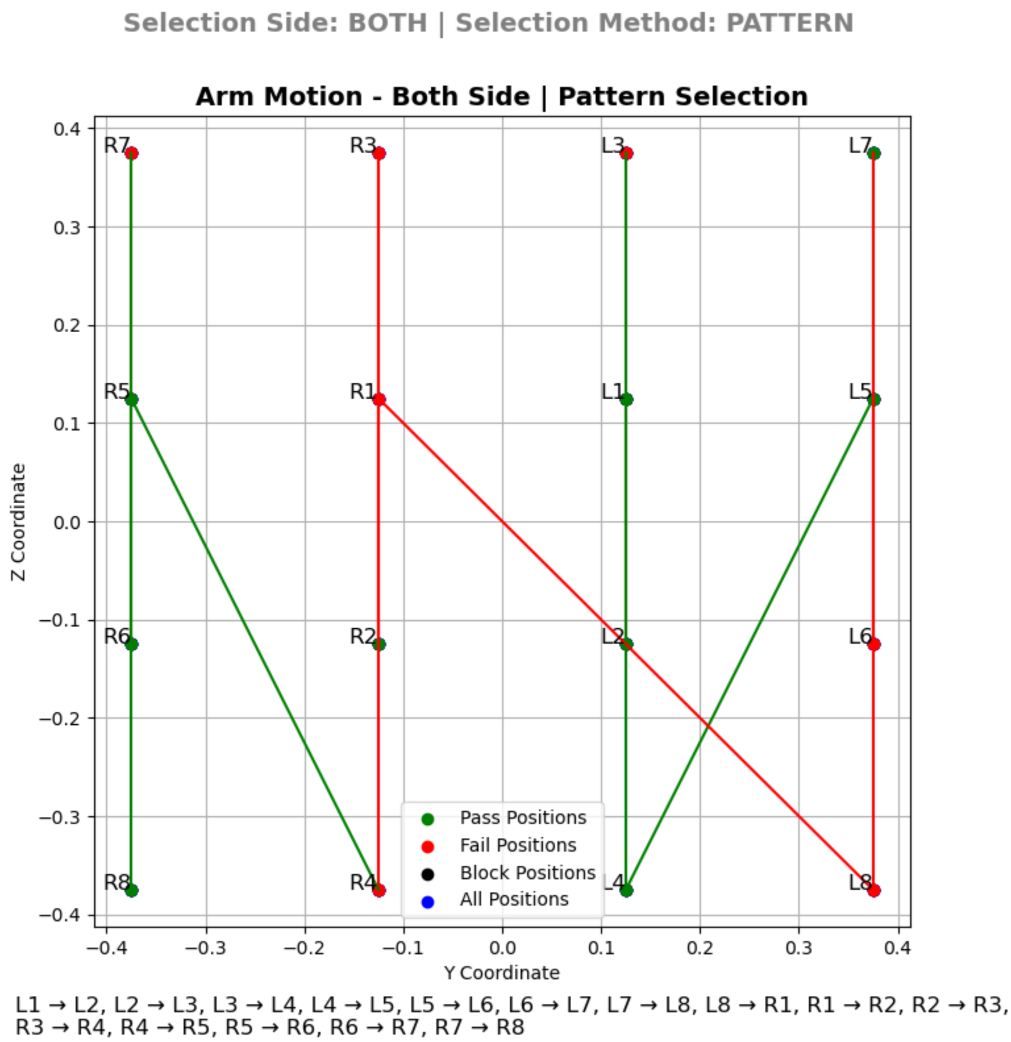


Figure 5.4: Arm motion test on Both-side in Pattern mode showing 16 target points (matrix size: 0.8m x 0.8m) [1]

Arm Motion with the Both-side test in Random

The robot's arm selects from all 16 target points randomly, moving alternately between left and right sides in no fixed order. In Figure 5.5 setup simulates a less predictable interaction, which can help analyze the patient's reflexes, coordination, and attention to varied stimuli during therapy.

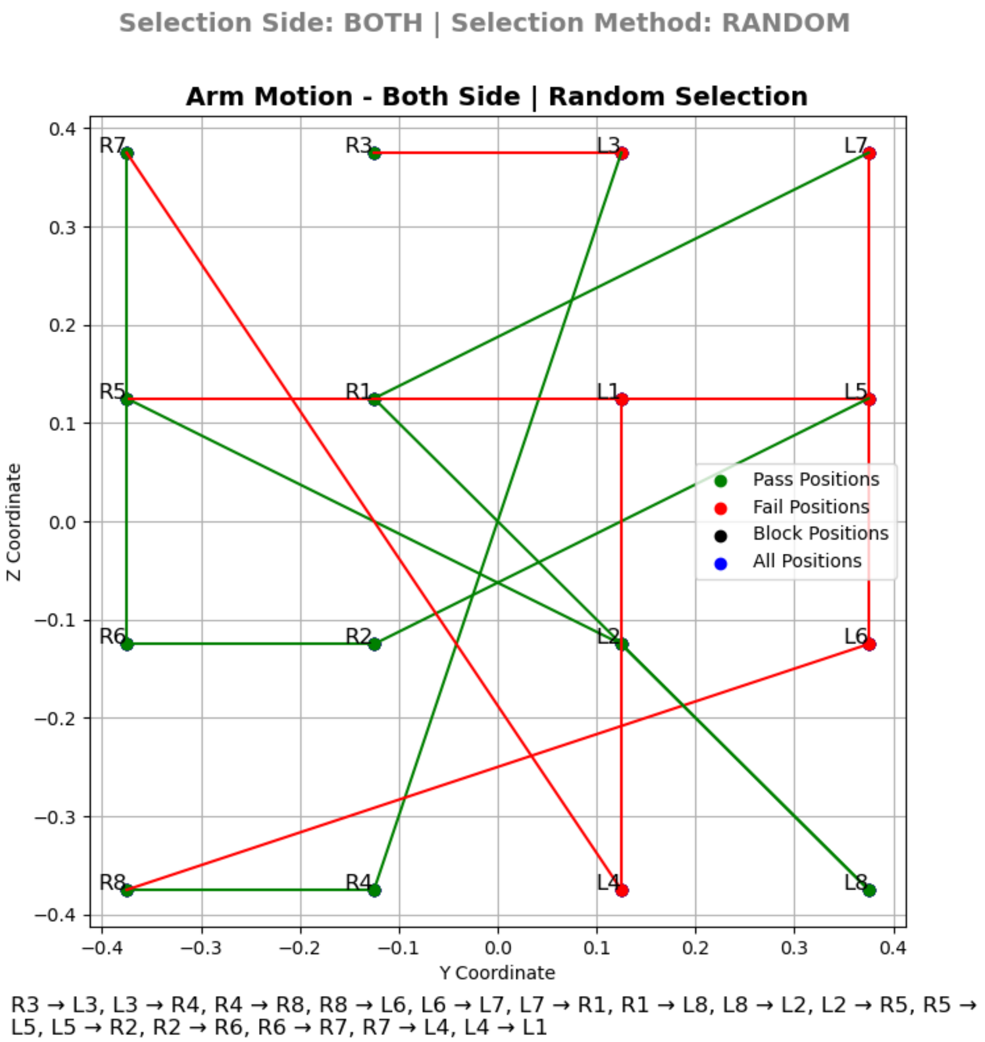


Figure 5.5: Arm motion test on Both-side in Random mode showing randomized target interaction [1]

Arm Motion with the Left-side test in Pattern

In Figure 5.6, the robot's arm targets only the 8 predefined points on the left side, following a structured and repeatable pattern. It focuses therapy on one half of the patient's reachable space useful for left-side motor rehabilitation or assessment of performance.

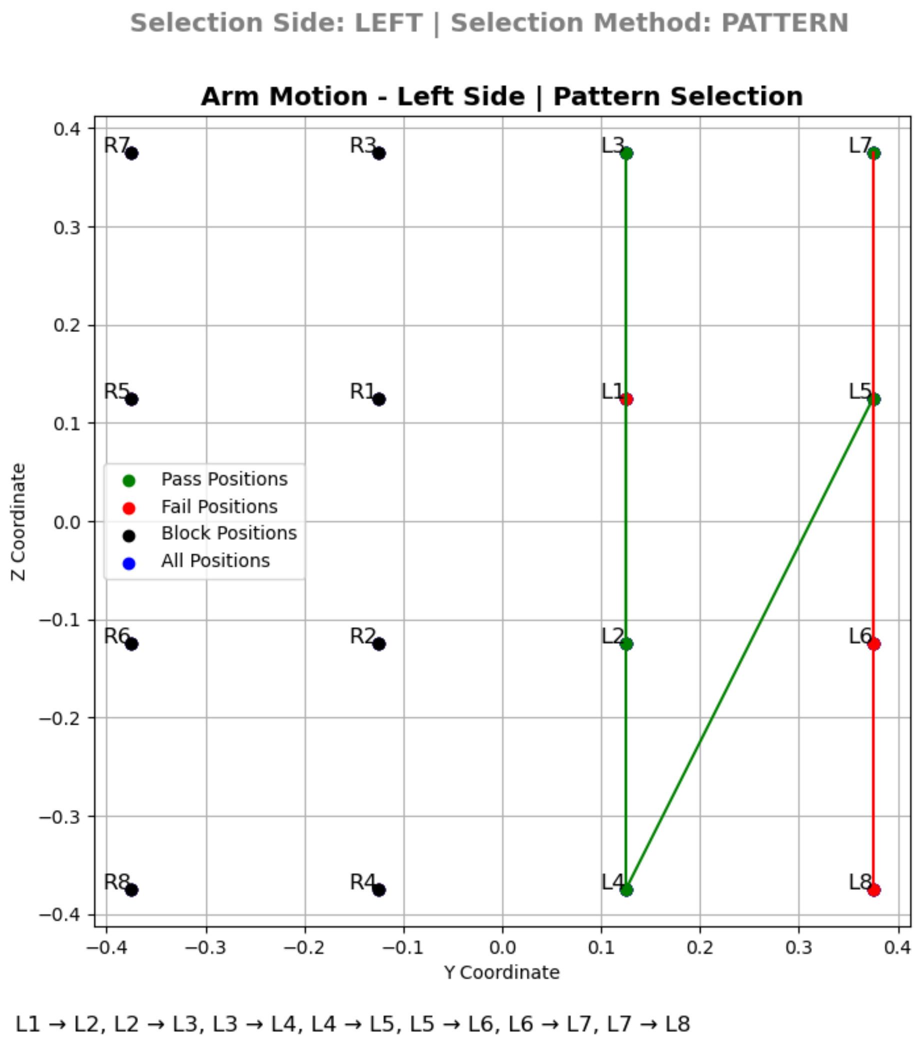


Figure 5.6: Arm motion test on Left-side in Pattern mode [1]

Arm Motion with the Left-side test in Random

In this variation, the arm performs random motions limited to the 8 left-side positions. It evaluates and shown in Figure 5.7 how the patient reacts to spontaneous arm movements toward their left, helping to identify any inconsistencies in attention or control when only one region is targeted.

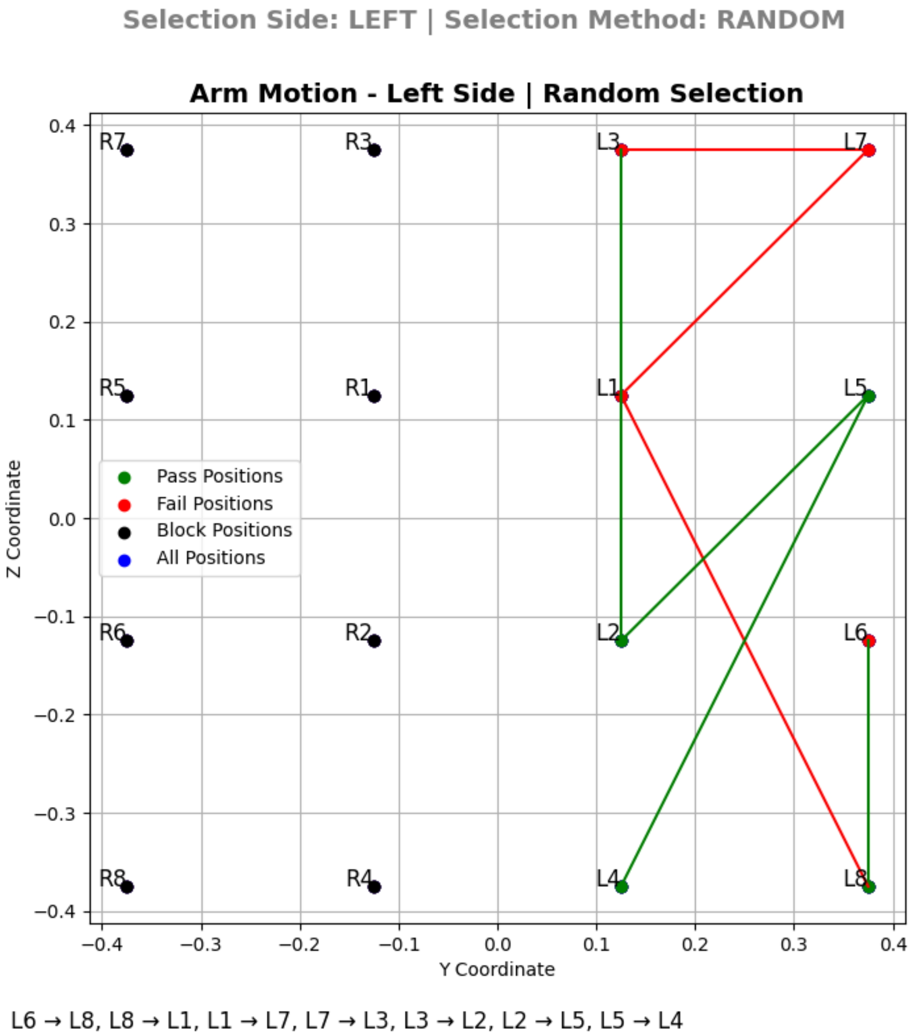


Figure 5.7: Arm motion test on Left-side in Random mode [1]

Arm Motion with the Right-side test in Pattern

Figure 5.8 test uses only the 8 motion points located on the right side. The robot's arm moves in a predetermined order, allowing for focused analysis on how well the patient follows a consistent right-sided motion routine.

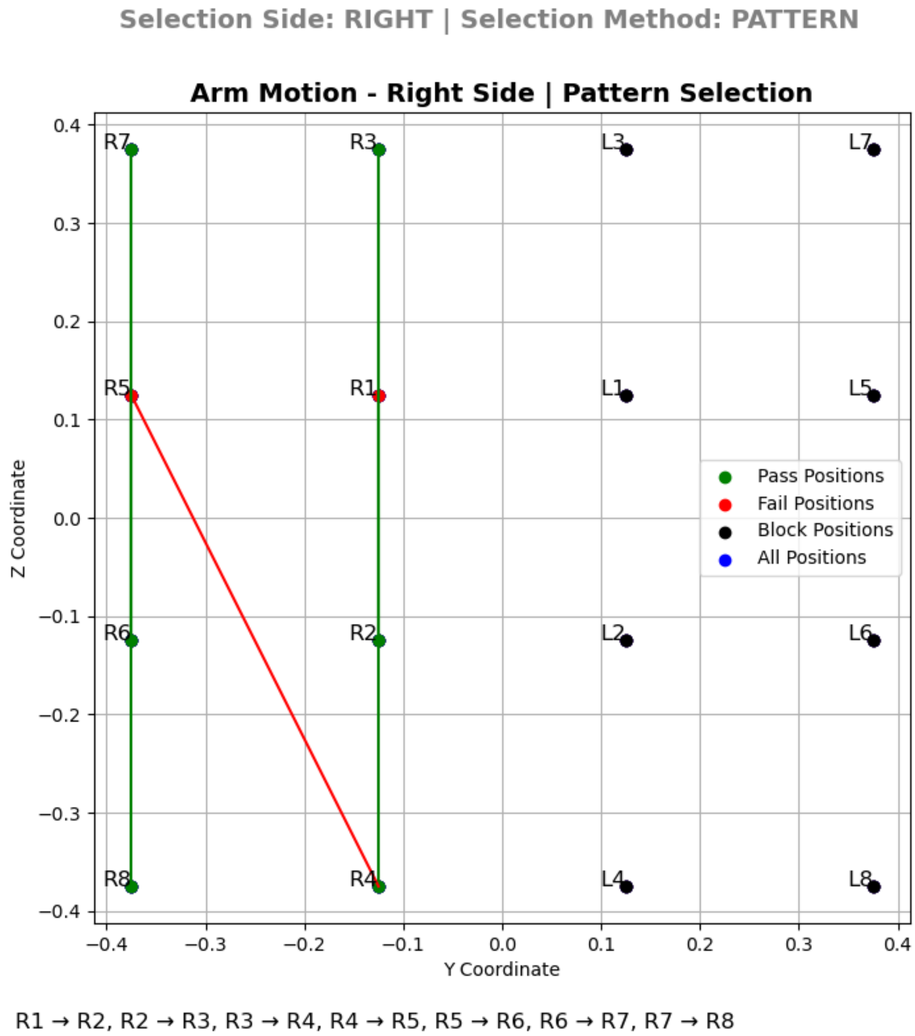


Figure 5.8: Arm motion test on Right-side in Pattern mode [1]

Arm Motion with the Right-side test in Random

The robot's arm performs randomized movements restricted to the right-side positions. Figure 5.9 setup is useful for observing spontaneous reaction and comfort level during unstructured movements focused only on the patient's right-hand side.

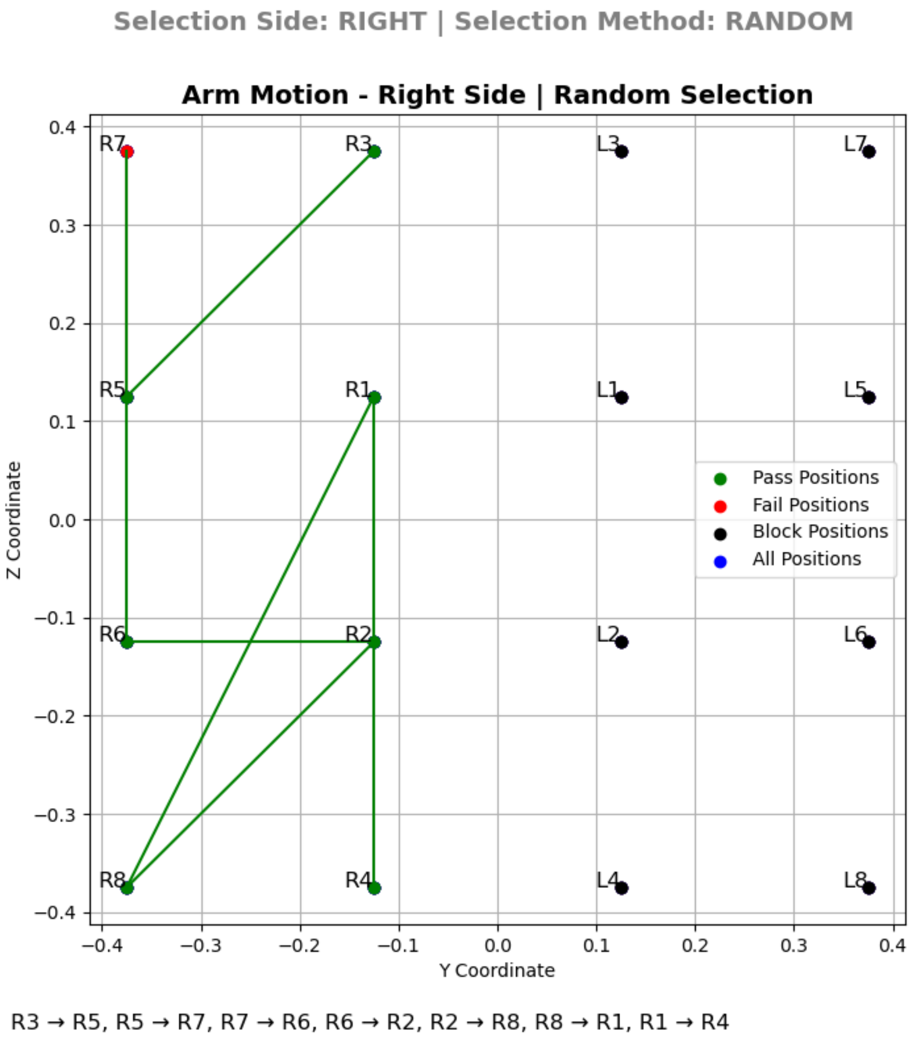


Figure 5.9: Arm motion test on Right-side in Random mode [1]

5.5 Arm Movement with Recipes

As seen in the picture below, this testing mode has 30 different trajectory visualizations that are grouped in a 5×6 matrix. Each cell in the matrix shows a different movement sequence that the robotic arm makes as it moves from one specified place to another. The purpose of these trajectories is to test patients' motor coordination, attentiveness, and visuospatial awareness. The number of waypoints determines how hard each movement is. Simple jobs only have two-way transitions, whereas more difficult motions include sequences of three or four points. Also, the end-effector's speed slowly rises as the levels go up, making the task harder as the patient becomes better.

An external database implemented as a spreadsheet file contains all of the movement sequences. This file is the main source for designing trajectories. These paths are sorted into five levels, each one indicating a different level of difficulty and matching the extent of hemispatial neglect seen in patients. There are five levels as we can see the reults in Figure 5.10 to Figure 5.14:

- **Level 5:** Short-distance moves between two places serve as the baseline and easiest level of difficulty.
- **Level 4:** Long-distance changes between two sites that need a lot of spatial awareness.
- **Level 3:** Long-distance sequences with three points add to the difficulty of navigation and timing.
- **Level 2:** Long-distance sequences with three points that include sharp or irregular turns raise cognitive stress since they are more complicated.
- **Level 1:** The hardest set of challenges, which include three- and four-point sequences with complicated changes in direction. These tasks are meant to test high-level coordination and attentiveness.

The testing system is meant to be flexible and tailored to each patient. When a new patient is assessed, a unique face ID and session record are created and saved in a small SQLite database. The idea of the system is based on older pattern-based and random testing modes, but it works better and smarter. Each movement is seen as a separate task, and gaze tracking and body posture analysis are used to keep an eye on how the patient is interacting with the robotic arm's end effector in real time.

Have check that the patient is looking at the end effector to make sure that each task is correct. The exercise is marked as successfully completed if the patient keeps their eyes on the movement the whole time. There are 30 of these tasks in a full session, and the number of successful completions is compared to a configurable threshold set by physicians based on the patient's ability and evaluation criteria to see if they can move on to the next level.

The system incorporates an intermediate adjustment period to make it easier to move between difficulty levels and to minimize sudden spikes in cognitive strain. In particular, the last 10 movements from the previous level are done again, but faster, before new movement patterns are added. This helps patients get used to the new environment slowly and reduces confusion, especially in people with cognitive flexibility issues.

For example, if a patient passes Levels 5 and 4 but fails Level 3, they have to do Level 3 again. If the patient fails again on the second try, the session is over, and the highest level they completed is recorded for that patient's face ID. When the patient comes back, they start testing again from the last level they passed, which lets them keep track of their progress and keep things going.

All the accompanying data, such as movement graphs, test results, audiovisual recordings, and metadata, is safely stored on a local drive. This large dataset is meant to be used not just for clinical evaluation but also for machine learning in the future. Continuous data gathering is very important at this stage because it is the first step in deploying the system. After collecting enough different types of patient data, machine learning models can be taught to find patterns in performance, spot unusual behavior, and tailor rehabilitation plans to each patient based on many types of input.

5.6 Patient Test Report Details

Every patient who took the exam got a thorough report card as part of the evaluation procedure. The report has the following information:

- **Name:** The patient's complete name who is taking the test.
- **Test Time:** The precise time and date when the test took place.
- **Test Number:** Each test session gets a unique number that can be used to keep track of and refer to it.

- **Selection Method:** The kind of test chosen, such random, patterned, or based on a set formula.
- **Minimum Passing Threshold (Count):** The number of tasks that must be passed at least to be considered a successful test attempt.
- **Minimum Passing Percentage:** The lowest number of correct attempts needed to achieve the success criteria.
- **Number of Passed Tasks:** The total number of test tasks that the patient successfully completed, with a maximum of 30.
- **Achieved Percentage:** This is the real percentage of successful attempts made by the patient, based on how well they did on the test.

Arm Motion with the Both-side Test Level 5 As Per Recipe

Figure 5.10 illustrates the arm movement trajectories assigned for Level 5. The motions are short and simple, consisting of two-point transitions across both sides, designed for initial patient engagement and baseline evaluation.

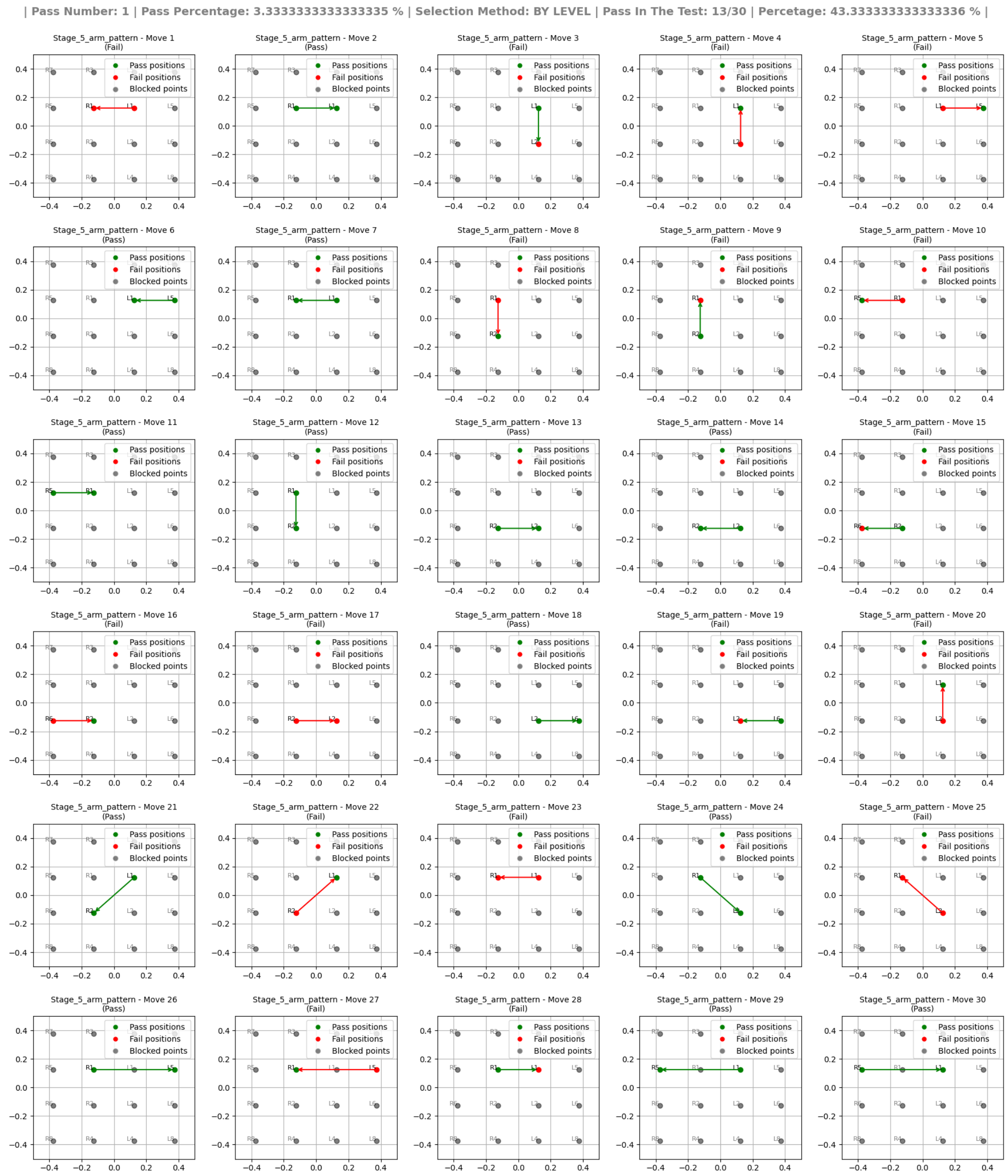


Figure 5.10: Level 5 test: Basic two-point trajectories with minimal reach distance [1]

Arm Motion with the Both-side Test Level 4 As Per Recipe

The Figure 5.11 displays Level 4 trajectories, which feature longer-distance transitions between two points, increasing the challenge while maintaining a relatively simple motion path.

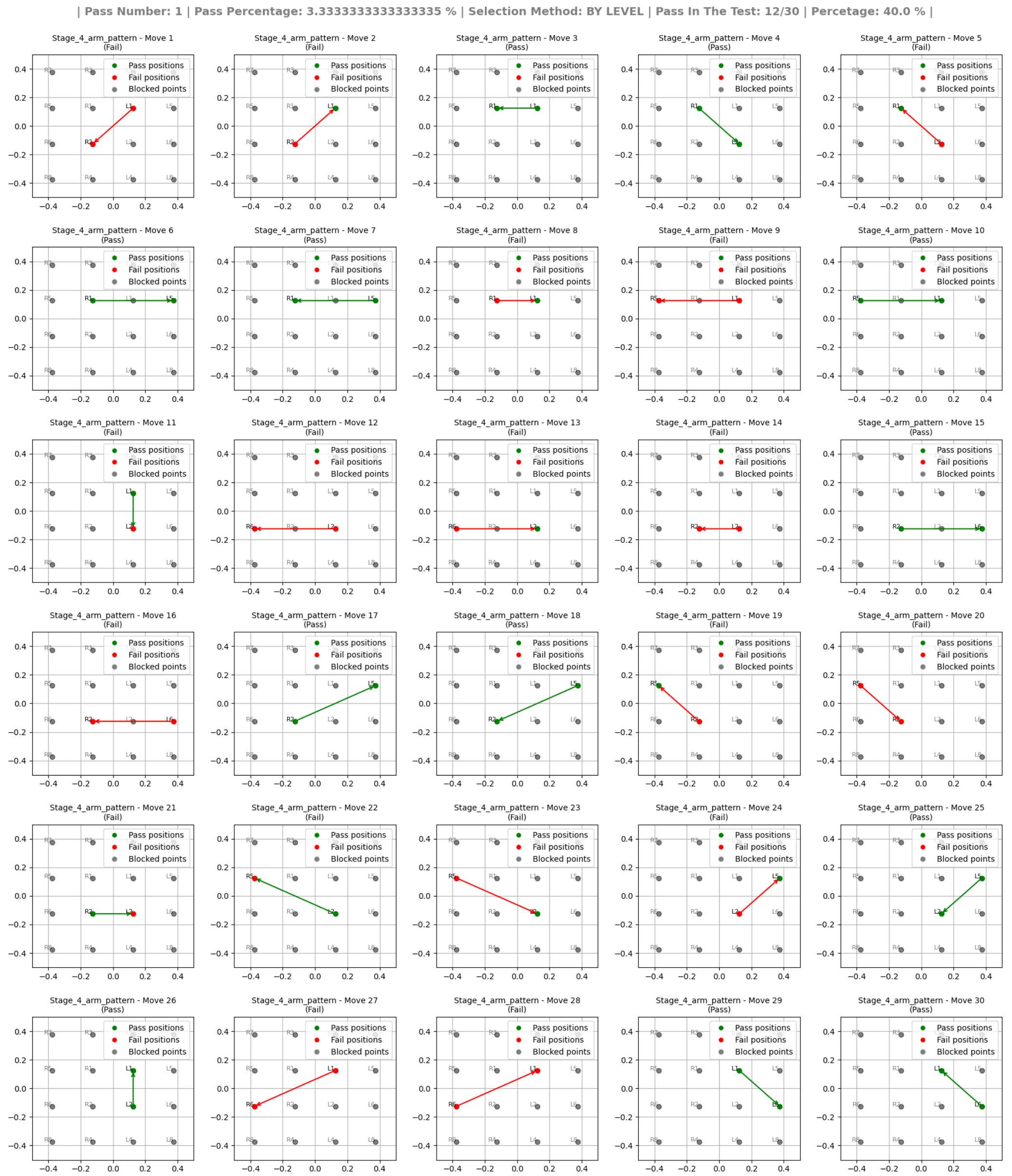


Figure 5.11: Level 4 test: Two-point movements with extended reach range [1]

Arm Motion with the Both-side Test Level 3 As Per Recipe

In Figure 5.12 the Level 3 motion paths, introducing sequences with three points. These trajectories are longer and include intermediate targets, making the motor task more demanding and suitable for evaluating mid-stage conditions.

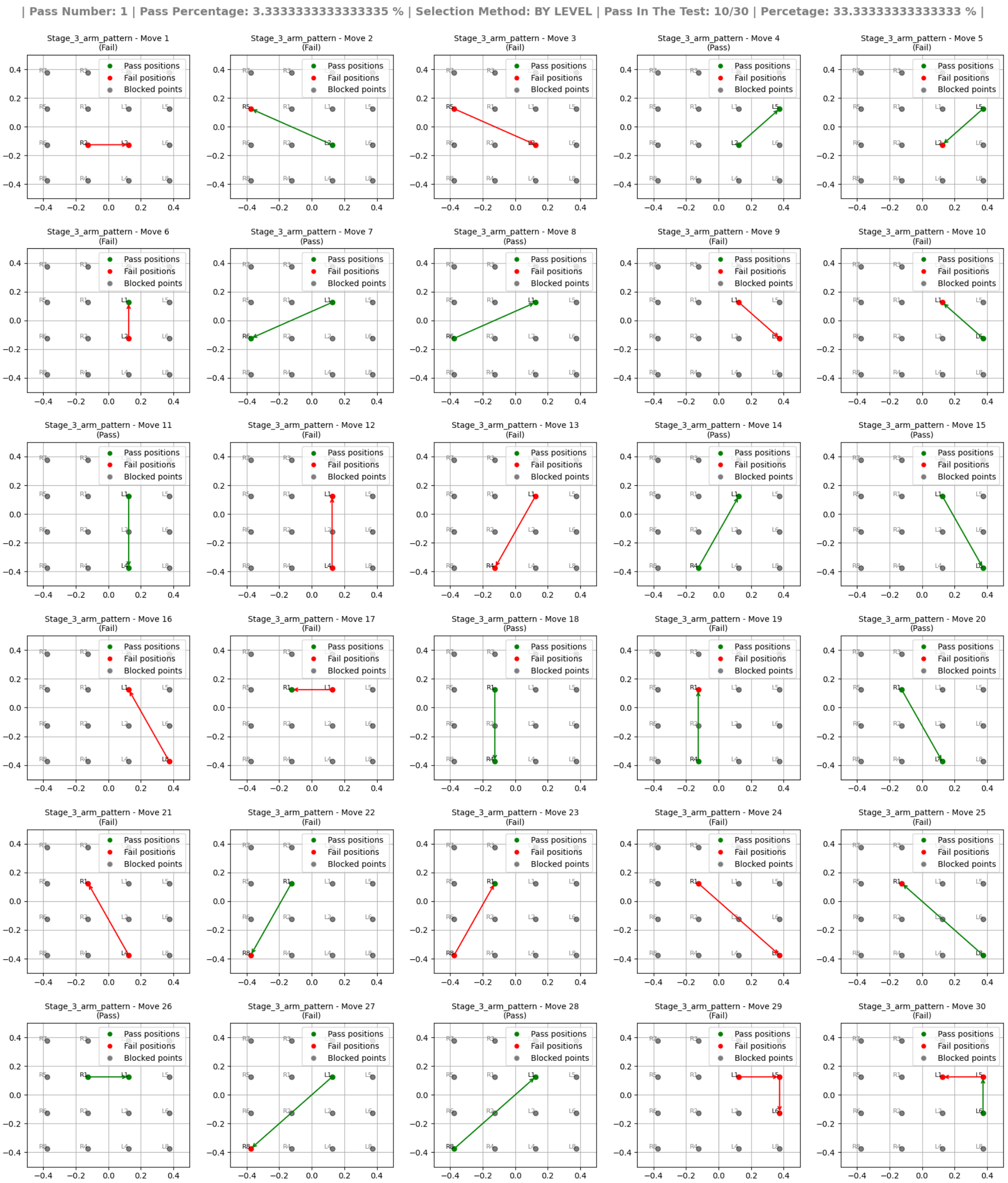


Figure 5.12: Level 3 test: Three-point trajectories with simple direction changes [1]

Arm Motion with the Both-side Test Level 2 As Per Recipe

In Level 2, depicted in Figure 5.13, the trajectories involve three points with more complex angular changes and distances. This increases the cognitive and motor demands on the patient during the assessment.

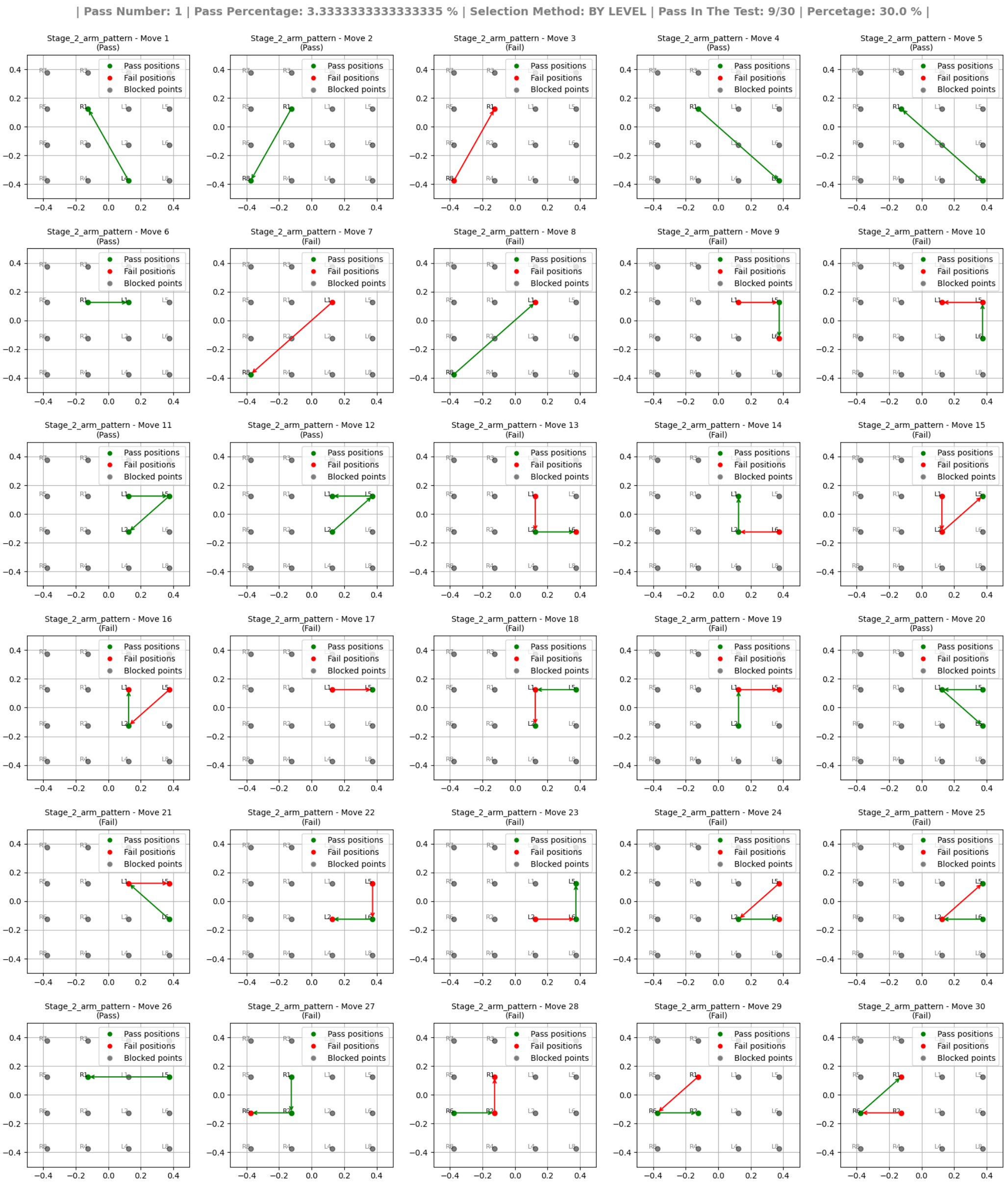


Figure 5.13: Level 2 test: Complex three-point movements with irregular spacing [1]

Arm Motion with the Both-side Test Level 1 As Per Recipe

Figure 5.14 presents the most challenging set of movements at Level 1, which includes both complex three-point and four-point trajectories. These patterns are designed to evaluate the highest level of motor and cognitive coordination.

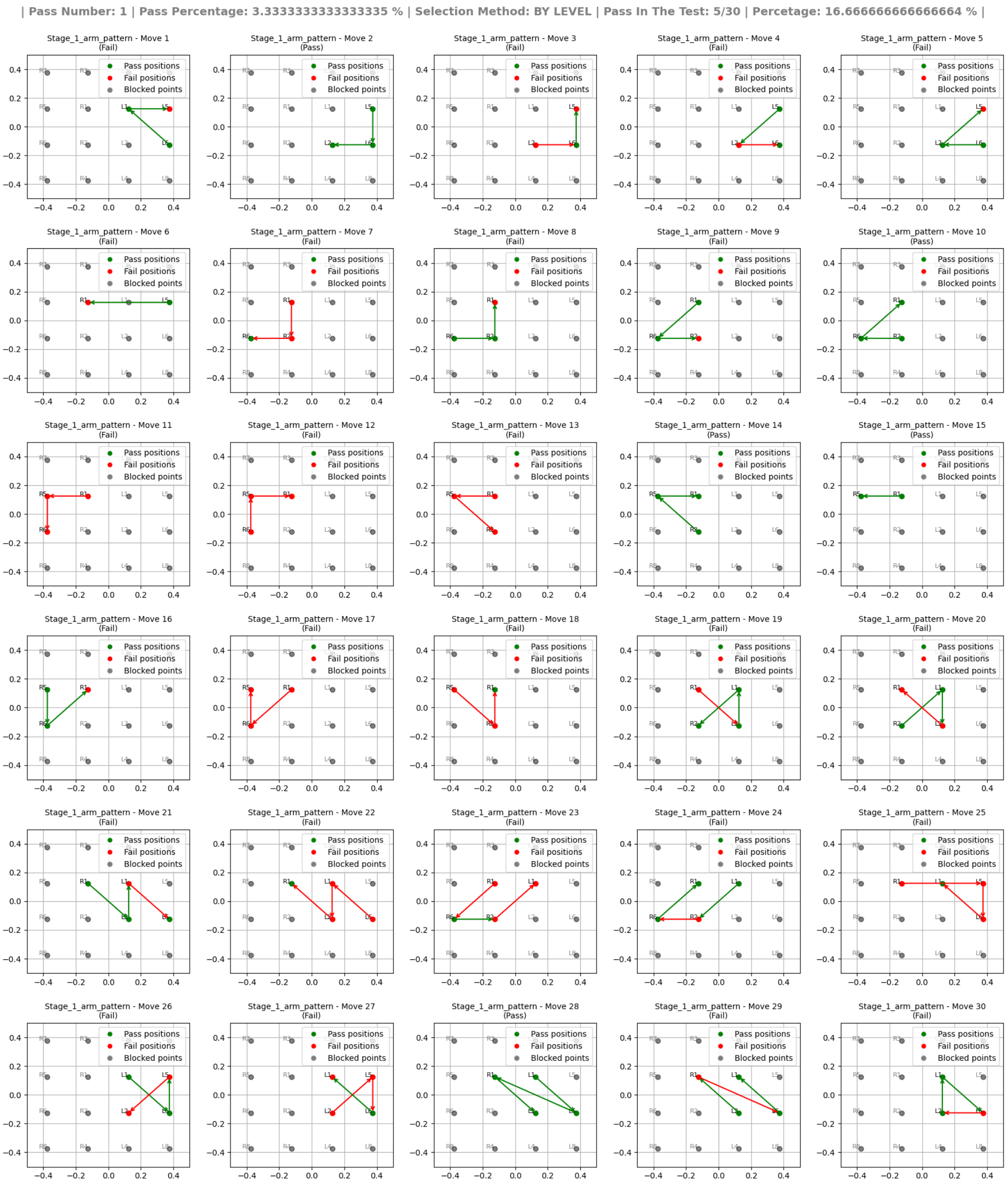


Figure 5.14: Level 1 test: Advanced multi-point trajectories with diverse directionality [1]

5.7 Comparative Analysis

Table 5.2 shows the proportion of accuracy for moves that were either planned out or random. It shows how many times the robot was able to reach the goal point relative to how many times it tried. This measurement is used to figure out how accurate the system is in getting to the right places.

Table 5.2: Success rates by test type

Test Mode	**Attempts**	**Success**	**Accuracy**	**Remarks**
Calibration Time	10	7	70%	Minor misalignments during setup
Random Arm	16	16	100%	High reliability in random motions
Pattern Arm	30	30	100%	Stable performance in guided paths
Circular Base	36	36	100%	Base control with precision

5.8 Error Analysis

Key observations:

1. Arm movements were consistent within ±10mm of their position.
2. Base Movement Navigation picked up a drift of about 5 cm over the whole circular journey.
3. It takes a long time to calibrate, and it should be done in a controlled environment where others can't become involved.
4. The Voks open source lightweight model can recognize 80% of voice commands in a lab setting (it can get better with Google speech detection).

5.9 Limitations

Although the system works well in controlled conditions, there are still a few limitations that affect its real-world performance. These include hardware constraints, system responsiveness, and environment related challenges.

1. Execution times are around 25% longer than those of human therapists, which could make sessions less efficient.

2. Not very good at adapting to changes in a patient's behavior or discomfort in real time, which could decrease the level of therapeutic involvement.

3. The programmed settings limit data capture and logging; actions or events that are not programmed may not be recorded.

4. Data must be analyzed and categorized right away because there isn't enough storage space on board. Also, storing data in a compressed format takes more time to compute, which could slow down other processes.

5. Battery life only lasts 5 to 6 hours of continuous use, which could make it hard to use in long therapy sessions or places where charging isn't simple to get to.

6. The facial recognition system is quite sensitive to the amount of light in the room, which can make it less accurate in places where the light changes or is low.

Chapter 6

Conclusions

"Computer Science is no more about computers than astronomy is about telescopes."

E. W. Dijkstra

This thesis has shown that a ROS2-based socially assistive robotic framework for psychological and motor rehabilitation works by employing the TIAGO robot as the main platform. The technology was designed to help adults whose brains had been damaged on one or both sides. It was supposed to give people tailored therapy routines by using preprogrammed robot behavior, real-time feedback, and several ways to connect.

The major purpose of this effort was to put into operation and test a number of therapy-specific modules, including as self-navigating systems, voice recognition, vision-based tracking, and robotic exercises to keep the pesians interested. The robot is ready to be tested in the Living Lab at the University of Pavia. Have used 2D LiDAR-based SLAM to execute navigation testing with pre-set test points that moved in circles and straight lines. The robot was able to follow 18-point and 9-point circular navigation patterns around the patient with very little drift or failure to find its way. This showed that the navigation stack was stable.

The tests using the robotic arm's movements were equally as extensive, and they covered both symmetric (pattern-based) and asymmetric (randomized) motion recipes. The goal of these activities was to make the damaged hemisphere's movements function better. Have set up the workspace with a 16-point arm grid and utilized ROS2 action interfaces to control the arm's movements. The recipes said that the system could achieve very accurate positions and reproduce them reliably over many attempts. Have also employed and evaluated level-based trajectory sequences, which can be as basic as moving between two points or as complicated as moving between five points. This proved how beneficial and adaptable recipe-based rehabilitation routines can be in a clinical context.

The Vosk engine lets the robot employ an offline voice recognition pipeline. This made it easy and safe to talk to the robot even when it wasn't linked to the internet. The robot also used its RGB-D camera to see the patient's face and posture in real time and follow them. This helped it determine out how interested they were. In the future, this visual data will help us to figure out how someone is feeling, how long they can pay attention.

Safety and durability is also things that were thought about when have decided to put this project into action. The control layer now had an emergency stop function, and fault recovery scripts made sure that the system could remain functioning smoothly even if there were little problems while it was doing the operation. During repeated tests, these measures functioned well to keep the system working and the patients safe.

The experimental validation results suggest that robotic devices that help people socially can be employed in real life for neurorehabilitation in adults. The system could run structured, multi-modal therapy sessions on its own, keep an eye on how the patients were doing in real time, and collect data from the sessions for subsequent analysis. There are still clinical trials with patients to come, but the results suggest that the platform can benefit rehabilitation experts by automating elements of therapy that are repetitive, measurable, and customizable.

This study lays the groundwork for the next generation of rehabilitation systems that will use robotics, cognitive science, and smart interaction. The framework that was made not only works technically and functionally, but it also fits the needs of neurorehabilitation, especially for adults with brain injuries on one side. Learning-based task adaptability, affective computing for emotional input, and telerehabilitation platforms for remote access are all things that could be added in future upgrades to make the experience more personal. This work does give us a proven and repeatable robotic system that really helps in the burgeoning field of intelligent rehabilitation technology.

6.1 Future Works

Future research will concentrate on improving the adaptability, intuitiveness, and accessibility of robotic neurorehabilitation systems, building on the foundation laid in this thesis. Crucial directives involve using reinforcement learning and AI-driven customisation to adapt therapy sessions dynamically according to patient progress and engagement. Incorporating powerful human-robot interaction techniques, such gaze tracking, facial expression detection, and natural language comprehension, would facilitate more sympathetic and engaging therapeutic experiences. The advancement of cloud-based tele-rehabilitation technologies would enable remote monitoring and data-informed modifications by therapists, hence enhancing access to therapy. Additionally, integrating multi-modal sensor fusion to analyze visual, audio, and physiological inputs (like heart rate and skin conductance) will yield enhanced understanding of patient conditions for more effective interventions. Research into AI-driven fault detection and predictive maintenance will enhance system reliability and patient safety. The implementation of augmented and virtual reality environments offers the potential to enhance the engagement and motivation of rehabilitation exercises. Ultimately, investigating collaborative multi-robot frameworks and clarifying AI methodologies will enhance the efficacy of therapy sessions and promote transparency in robotic decision making. These developments will enhance intelligent rehabilitation technology, making it more customized, accessible, and successful.

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